

Econ 102B

Applied Econometrics

Stanford University

Course Materials for Week 4 through Week 6

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Week 4 through Week 6

Goals:

1. Understanding how to calculate confidence intervals and conduct hypothesis tests on the regression coefficients β_1 and β_2 .
2. Understanding how to calculate confidence intervals and conduct hypothesis tests on the expected value of Y, for a given X-value.
3. Becoming acquainted with dummy variables (or indicator variables) in a regression.
4. Learning how dummy variables can be used in a simple linear regression to run two-population hypothesis tests.
5. Becoming acquainted with prediction intervals and how they differ from confidence intervals.
6. Learning how to compute an R^2 -value and understanding its relation to a correlation coefficient.
7. Gaining familiarity with some nonlinear functional forms.

Weeks 4 through 6 Worksheet

1. Suppose we wish to predict the amount a person spends on a new car based on their annual income. A sample among 4 individuals provides the following:

Income (000)	Car Price (000)
60	24
83	40
52	28
76	35

- (a) Determine the OLS regression line for the above sample.
- (b) Assume the regression line in part (a) conforms to the true population line. Estimate the (constant) variance on car price for any given income level. Then, determine the probability a person whose annual salary is \$65,000 spends \$36,000 or less on a new car.
- (c) Based on the sample, is there evidence that income is a viable predictor of car price? That is, does there appear to be a correlation between the variables? Conduct the appropriate hypothesis test using $\alpha = 5\%$. Determine the associated p-value.
- (d) Use the sample to develop an 80% confidence interval for β_2 .
2. Consider a researcher who wishes to study the amount of money a person earns annually. The August 2021 Current Population Survey (CPS) collected (among other things) data on a person's annual earnings along with their education level (i.e., their number of completed years of education). The survey participants serve as a sample for all wage earners within the United States. Consider the model:

$$\text{Earnings} = \beta_1 + \beta_2 \text{ Education} + \varepsilon$$

where 'Earnings' represents one's personal annual income (in thousands) and 'Education' represents one's number of completed years of education.

This exercise is based on a sample of 509 participants in the CPS survey.

- (a) The researcher believes that American workers earn, on average, more than \$5,200 annually for each additional year spent in education. Conduct a hypothesis test, at $\alpha = 5\%$, on whether the data provides ample evidence to support this belief.

- (b) The researcher further believes that the average earnings among all Americans with 16 years of education is less than \$65,000. Conduct a hypothesis test, at $\alpha = 5\%$, on whether the data provides ample evidence to support this belief.
3. A study was conducted in Tennessee in which a student cohort was followed and observed from kindergarten through third grade. Students were randomly assigned either, (1) small classes, (2) regular-sized classes without a teacher's aide, or (3) regular-sized classes with a teacher's aide. The study hoped to analyze whether students perform better on tests in a smaller class setting and/or whether they perform better within classes having a teacher's aide.

In this exercise we will only contrast students within small classes versus regular-sized classes. Below is a table of data on test scores from 62 students from small classes and 40 students from regular-sized classes:

Small Class Scores				Regular Class Scores	
996	843	844		862	921
920	942	963		934	904
983	857	900		969	989
900	858	960		1008	858
1043	1060	927		857	867
1049	916	999		855	975
1079	991	892		927	837
989	1080	986		1071	863
931	1054	1079		881	999
887	891	940		957	943
855	962	985		886	1020
913	947	882		997	877
917	819	854		942	998
914	962	896		811	834
1020	811	864		879	857
857	1044	939		924	908
922	1012	893		861	927
848	867	948		832	895
884	847	890		837	909
856	923	932		887	
925	873			1052	

Use regression to run a two-population hypothesis test on whether the two groups have equal test averages. Verbalize the physical meaning of b_1 , b_2 , mean sum of squared errors, $\text{Var}(b_1)$ and $\text{Var}(b_2)$ within your regression results.

4. Reconsider the August 2021 Current Population Survey (CPS) discussed in Question 2. The survey also collected information on the number of hours a person works in a particular week. Here, we consider the model:

$$\text{Earnings} = \beta_1 + \beta_2 \text{HrsWork} + \varepsilon$$

where ‘Earnings’ represents one’s personal annual income (in thousands) and ‘HrsWork’ represents one’s number of weekly hours worked.

This exercise is based on a sample of 424 participants in the CPS survey.

- (a) Determine the 70% interval on the average earnings among all United States wage earners who work 40 hours per week.
- (b) Reid Andrews is employed in Lincoln, Nebraska and works 40 hours per week. Determine the 70% interval on Reid’s annual earnings. (Note: from the Stata output, we find $\text{MSE} = 1717.63$.)

(Note: Due to time constraints, this exercise is typically not presented in lecture. However, the solution to this exercise is provided on Page 6 of this handout.)

5. Consider a sample of $n = 4$ cities in the United States, consisting of each city’s size (in square miles) along with each city’s population (in thousands). Specifically, we have:

City	SQMiles	Population (000)
City 1	37	115
City 2	25	74
City 3	89	225
City 4	162	200

We consider the model:

$$\text{Population} = \beta_1 + \beta_2 \text{SQMiles} + \varepsilon$$

When applying the sample to the above model, determine the R^2 -value of the ensuing OLS regression line.

6. For the population of chief executive officers (CEOs), let Y = the CEO's annual salary (in thousands) and let X = annual firm revenue (in millions). In an effort to analyze the relationship between revenue and CEO salary, *Business Week* collected data on 209 CEO salaries and their respective firm's annual revenue.¹

A subset of the entire data set is as follows:

Annual Revenue (X) (millions)	CEO's Salary (Y) (thousands)
175.2	453
596.8	567
1651.9	815
2518.6	991
6125.9	1122
13538	1368
17453.8	1620
20604.0	3844

You further recall reading an article in *The Economist* which asserted that large companies try to roughly pay their CEOs in proportionate percentages. That is, for every 1% rise in revenue the CEO's salary increases by some constant $k\%$. In other words, the salary-revenue relationship approximately exhibits constant elasticity. You would like your regression model to include this assumption and render an estimate of k .

Develop an appropriate regression model which predicts a CEO's salary as a function of the company's annual revenue. Which functional form should we use to adhere to the 'constant elasticity' assumption?

7. You have just received a job offer from a consulting firm in San Francisco. However, there has yet been no discussion of your starting salary and you are meeting with the firm tomorrow for this discussion. At present you are not sure what would constitute a lowball offer versus a generous offer.

With regard to salary standards in the consulting industry, you believe the following:

- (i) Because of the competitive nature of the industry, just about all newly-hired consultants working in San Francisco earn roughly the same starting salary (although you do not know the precise figure).

¹ Adapted from Wooldridge, Jeffrey M., *Introductory Econometrics: A Modern Approach*, 4th Edition, South-Western, 2009.

- (ii) Like many industries, consultants are given a constant-percentage raise each year. That is, if a consultant makes \$80,000 one year, then the following year that consultant will make $\$80,000(1 + r)$ for some constant rate, r , whose precise value you do not know.

In an effort to estimate the starting salary and yearly raise percentage, you gather data on seven different San Francisco consultants regarding the length of time they have been in the profession and their current salary. The data is given by:

Consultant	Years in Profession (X)	Current Salary (Y)
Consultant 1	3	76,500
Consultant 2	5	87,800
Consultant 3	5	95,600
Consultant 4	6	101,200
Consultant 5	8	111,700
Consultant 6	10	117,900
Consultant 7	11	133,100

- (a) Determine the proper functional form for the regression so as to estimate a starting salary offer as well as the yearly percentage raise in salary.
- (b) Based on your model from part (a), estimate the salary you will be offered in tomorrow's meeting. Also, if you ask the firm about their policy on annual raises, what annual percentage raise do you expect them to provide?
- (c) Suppose the X-variable is measured in months instead of years. Further, suppose the Y-variable is measured in thousands of dollars. Determine how the regression line found in part (a) would change. Intuitively, how should b_1 change? How should b_2 change?

Worksheet Question 4 Solution

The regression line for the sample (of size $n = 424$) is given by:

$$\text{Earnings} = -1.291 + 1.349 \text{ HrsWork}$$

Therefore, whether we calculate a confidence interval on the average earnings among *all* workers having $\text{HrsWork} = 40$, or whether we calculate a prediction interval on the earnings of *a particular* worker having $\text{HrsWork} = 40$, the center of the interval is given by:

$$-1.291 + 1.349 (40) = 52.669$$

- (a) This part asks about the population average among all earners who work a 40-hour work week. Since this is a calculation on an entire group of people, we seek a **confidence interval**.

From above, our ‘best guess’ of the average is 52.669

The variance is given by:

$$\text{Var}(\hat{Y}) = \text{Var}(b_1) + (40)^2 \text{Var}(b_2) + 2(40) \text{Cov}(b_1, b_2) = 4.06$$

Note: The calculations for $\text{Var}(b_1)$, $\text{Var}(b_2)$ and $\text{Cov}(b_1, b_2)$ are found through Stata’s ‘vce’ command.

We further have $\#SD = \text{TINV}(.30, 422) = 1.038$, which leads to the 70% confidence interval being:

$$52.669 \pm (1.038) (\sqrt{4.06}) \Rightarrow [50.577 \text{ up to } 54.761]$$

- (b) This part asks for the annual earnings of a *particular worker* who works a 40-hour work week. Since this is a calculation on a specific member of the population, we seek a **prediction interval**.

From above, our ‘best guess’ of a particular worker’s earnings is 52.669

The variance is given by:

$$\text{Var}(Y_{\text{new}}) + \text{Var}(\hat{Y}) = \text{MSE} + \text{Var}(\hat{Y}) = 1717.63 + 4.06 = 1721.69$$

We still have $\#SD = \text{TINV}(.30, 422) = 1.038$, which leads to the 70% prediction interval being:

$$52.669 \pm (1.038) (\sqrt{1721.69}) \Rightarrow [9.599 \text{ up to } 95.739]$$

Elaboration of Worksheet Question 5

A table showing various computations for the sample observations is as follows:

X	Y	$\hat{Y}$	$Y - \bar{Y}$	$\hat{Y} - \bar{Y}$	Residual	$(\hat{Y} - \bar{Y}) * (\text{Residual})$
37	115	116.43	- 38.5	- 37.07	- 1.43	53.01
25	74	105.65	- 79.5	- 47.85	- 31.65	1514.39
89	225	163.16	71.5	9.66	61.84	597.41
162	200	228.76	46.5	75.26	- 28.76	- 2164.81
	$\bar{Y} = 153.5$	$E(\hat{Y}) = 153.5$				Sum = 0

Notice that the sum of all $(\hat{Y} - \bar{Y}) * (\text{Residual})$ terms equals zero!

Notice that, for any particular observation:

$$(Y - \bar{Y}) = (Y + \hat{Y} - \hat{Y} - \bar{Y}) = (\hat{Y} - \bar{Y}) + (Y - \hat{Y}) = (\hat{Y} - \bar{Y}) + \text{Residual}$$

$$\Rightarrow (Y - \bar{Y})^2 = [(\hat{Y} - \bar{Y}) + \text{Residual}]^2$$

$$\Rightarrow (Y - \bar{Y})^2 = (\hat{Y} - \bar{Y})^2 + \text{Residual}^2 + 2 * (\hat{Y} - \bar{Y}) * (\text{Residual})$$

Summing across all observations then gives:

$$\Rightarrow \sum (Y - \bar{Y})^2 = \sum (\hat{Y} - \bar{Y})^2 + \sum \text{Residual}^2 + 2 \sum (\hat{Y} - \bar{Y}) * (\text{Residual})$$

$$\Rightarrow \sum (Y - \bar{Y})^2 = \sum (\hat{Y} - \bar{Y})^2 + \sum \text{Residual}^2 + 2 (0) \text{ (see last column in table)}$$

$$\Rightarrow \sum (Y - \bar{Y})^2 = \sum (\hat{Y} - \bar{Y})^2 + \sum \text{Residual}^2$$

$$\Rightarrow \text{SST} = \text{SSR} + \text{SSE}$$

Overall, in general, we will always have the relationship: $\text{SST} = \text{SSR} + \text{SSE}$

Elaboration of Worksheet Question 6

The table showing the original and transformed variables is given by:

X	Y	ln(X)	ln(Y)	ln(X)*ln(Y)
175.2	453	5.166	6.116	31.594
596.8	567	6.392	6.340	40.525
1651.9	815	7.410	6.703	49.668
2518.6	991	7.831	6.899	54.027
6125.9	1122	8.720	7.023	61.241
13538	1368	9.513	7.221	68.696
17453.8	1620	9.767	7.390	72.182
20604.0	3844	9.933	8.254	81.992
		Mean = 8.092	Mean = 6.993	Mean = 57.491
		Var = 2.575		

Using the typical OLS regression estimators (but applying them to the columns ln(X) and ln(Y) instead of X and Y) we have:

$$b_2 = \frac{\text{Cov}(\ln(X), \ln(Y))}{\text{Var}(\ln(X))} = \frac{57.491 - (8.092)(6.993)}{2.575} = 0.351$$

$$b_1 = \overline{\ln(Y)} - b_2 \overline{\ln(X)} = 6.993 - (0.351)(8.092) = 4.153$$

Therefore, the regression line is: $\ln(\text{Salary}) = 0.351 \ln(\text{Revenue}) + 4.153$

and, since the slope coefficient represents the constant elasticity, we estimate that a 1% rise in revenue will increase a CEO's salary by 0.351% (i.e., when revenue increases by 3% the CEO's salary increases a little more than 1%).

Elaboration of Worksheet Question 7, part (a)

The table showing the original and transformed variable is given by:

X	Y	ln(Y)	X*ln(Y)
3	76,500	11.245	33.735
5	87,800	11.383	56.914
5	95,600	11.468	57.340
6	101,200	11.525	69.149
8	111,700	11.624	92.989
10	117,900	11.678	116.776
11	133,100	11.799	129.787
Mean = 6.857		Mean = 11.532	Mean = 79.527
Var = 7.265			

Using the typical OLS regression estimators (but applying them to the columns X and ln(Y) instead of X and Y) we have:

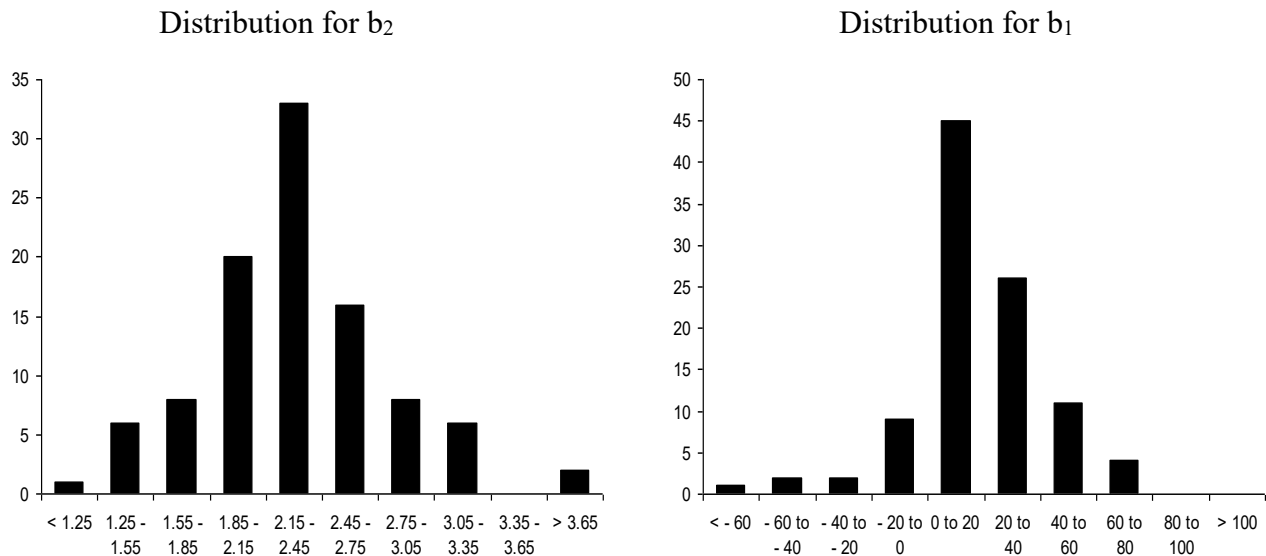
$$b_2 = \frac{\text{Cov}(X, \ln(Y))}{\text{Var}(X)} = \frac{79.527 - (6.857)(11.532)}{7.265} = 0.062$$

$$b_1 = \overline{\ln(Y)} - b_2 \bar{X} = 11.532 - (0.062)(6.857) = 11.103$$

Therefore, the regression line is: $\ln(\text{Salary}) = 0.062 (\text{Years}) + 11.103$.

Confidence Intervals and Hypothesis Tests on β_1 and β_2

We have previously discussed how the estimates b_1 and b_2 are random variables. For two separate samples, chances are that both b_1 and b_2 will differ depending on which sample we use. In a previous handout we simulated 100 different samples and calculated b_1 and b_2 for each separate sample. Ultimately, the distributions on b_2 and b_1 were as follows:



If we were to increase the number of samples beyond $n = 100$ to, say, $n = 10,000$ what do you think would happen to the shapes of the above distributions? Perhaps it is not a stretch to imagine that, if we were to add many, many more observations to the above distributions, both of them would more and more mimic a normal distribution. Indeed, under the assumptions of simple linear regression, the distributions for both b_1 and b_2 are normally distributed.

Further, we have already seen explicit formulas for the variances (and covariance) of these variables (see Handout #25). Since we are dealing with normal distributions having means and variances which we can estimate, we are now able to determine confidence intervals on the accuracy of these estimates as well as conduct hypothesis tests on potential values of the true parameter values β_1 and β_2 .

Stata Output

By now it is becoming apparent that there are a variety of formulas and calculations which underpin simple linear regression. The basic regression output generated by Stata contains (directly or indirectly) virtually all the calculations discussed in previous handouts. This handout is meant to specifically coordinate basic Stata output with the formulas and calculations we have previously seen.

As a working example consider a group of university students who are on the verge of graduating. The Admissions Office within the school wonders about the relationship between a student's graduating GPA and the SAT mathematics score the student achieved upon initially entering the school. The data for eight of the school's students are:

SAT Math Score (X)	Graduating GPA (Y)
640	3.4
700	3.2
550	3.5
540	2.8
620	3.0
490	2.4
710	3.6
720	3.8

The Stata output for the model, $GPA = \beta_1 + \beta_2 SAT + \varepsilon$ is shown atop the following page. The calculations we have covered thus far in class are circled and elaborated upon in the discussion after the output. Would you be able to reproduce this Stata output by hand? Do you understand what each circled number in the output represents?

The Stata output is given by:

Source	SS	df	MS
Model	.869719965	1	.869719965
Residual	.619029824	6	.103171637
Total	1.48874979	7	.212678541

GPA	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
SAT	.0040476	.0013941	2.90	0.027	.0006364	.0074587
_cons	.6979517	.8734786	0.80	0.455	-1.439373	2.835277

Number of obs = 8

F(1, 6) = 8.43

Prob > F = 0.0272

R-squared = 0.5842

Adj R-squared = 0.5149

Root MSE = .3212

Elaboration on the above calculations:

1. Since we assembled information on 8 students in the original sample, the sample size (i.e., the number of observations) is 8.

2. The estimate b_2 is given by $b_2 = \frac{Cov(X, Y)}{Var(X)}$. Here, we have:

$$E(XY) = \frac{1}{8}[(640)(3.4) + (700)(3.2) + (550)(3.5) + (540)(2.8) + (620)(3.0) + (490)(2.4) + (710)(3.6) + (720)(3.8)] = 2022.625$$

$$E(X) = \frac{1}{8}(640 + 700 + 550 + 540 + 620 + 490 + 710 + 720) = 621.25$$

$$E(Y) = \frac{1}{8}(3.4 + 3.2 + 3.5 + 2.8 + 3.0 + 2.4 + 3.6 + 3.8) = 3.2125$$

$$Var(X) = \frac{1}{8}(640^2 + 700^2 + 550^2 + 540^2 + 620^2 + 490^2 + 710^2 + 720^2) - (621.25)^2 = 6635.938$$

$$\text{So, } b_2 = \frac{2022.625 - (621.25)(3.2125)}{6635.938} = .0040476$$

The estimate b_1 is given by $b_1 = E(Y) - E(X) b_2$, which renders:

$$b_1 = 3.2125 - (621.25)(.0040476) = 0.6979517.$$

3. A table showing the original data, the predicted value of Y, the residuals and the associated squared residuals is given by:

X	Y	$\hat{Y}$	Residual	Residual ²
640	3.4	3.288	0.1116	0.0125
700	3.2	3.531	- 0.3313	0.1097
550	3.5	2.924	0.5759	0.3316
540	2.8	2.884	- 0.0837	0.0070
620	3.0	3.207	- 0.2075	0.0430
490	2.4	2.681	- 0.2813	0.0791
710	3.6	3.572	0.0283	0.0008
720	3.8	3.612	0.1878	0.0353

The sum of squared residuals is given by:

$$\sum_i \varepsilon_i^2 = .0125 + .1097 + .3316 + .0070 + .0430 + .0791 + .0008 + .0353 = .6190.$$

4. Since the sample size is $n = 8$, the degrees of freedom $= n - 2 = 6$. Therefore, the mean sum of squared residuals (i.e., MSE) is:

$$\frac{1}{N-2} \sum_i \varepsilon_i^2 = \frac{.6190}{6} = .1032.$$

5. Using the information from the previous calculation, the root mean squared error (or standard error of the regression) is:

$$\sqrt{\frac{1}{N-2} \sum_i \varepsilon_i^2} = \sqrt{.1032} = .3212.$$

6. The variance of b_2 is given by:

$$\text{Var}(b_2) = \frac{\sigma^2}{\sum_i (X_i - \bar{X})^2}$$

We estimate σ^2 as $\text{MSE} = \frac{1}{N-2} \sum_i \varepsilon_i^2 = .1032$, and we further have:

$$\sum_i X_i^2 = 640^2 + 700^2 + 550^2 + 540^2 + 620^2 + 490^2 + 710^2 + 720^2 = 3,140,700$$

$$\sum_i (X_i - \bar{X})^2 = 8 * 6635.938 = 53,087.50$$

$$\text{So, our estimated variance of } b_2 = .1032 \frac{1}{53,087.50} = 0.000001944$$

and the estimated standard deviation of b_2 is therefore $\sqrt{0.000001944} = 0.001394$.

The variance of b_1 is given by:

$$\text{Var}(b_1) = \sigma^2 \left[\frac{\sum_i X_i^2}{N \sum_i (X_i - \bar{X})^2} \right]$$

We estimate σ^2 as $\text{MSE} = \frac{1}{N-2} \sum_i \varepsilon_i^2 = .1032$, and we further have:

$$\sum_i X_i^2 = 640^2 + 700^2 + 550^2 + 540^2 + 620^2 + 490^2 + 710^2 + 720^2 = 3,140,700$$

$$\sum_i (X_i - \bar{X})^2 = 8 * 6635.938 = 53,087.50$$

$$\text{So, our estimated variance of } b_1 = .1032 \frac{3,140,700}{(8)(53,087.50)} = 0.7632$$

and the estimated standard deviation of b_1 is therefore $\sqrt{0.76317} = 0.8735$.

7. Since $n = 8$, the appropriate number of standard deviations for the 95% confidence interval on either b_1 or b_2 is $\text{TINV}(.05, 6) = 2.447$.

The 95% confidence interval on b_2 is then $.0040476 \pm (2.447)(0.001394)$

$$\Rightarrow .0040476 \pm .0034111 \Rightarrow .000636 \text{ up to } .007459$$

The 95% confidence interval on b_1 is then $0.6979517 \pm (2.447)(0.8735)$

$$\Rightarrow 0.6979517 \pm 2.137325 \Rightarrow -1.439373 \text{ up to } 2.835277$$

8. The hypothesis testing setup on whether the slope coefficient is 0 is given by $H_0: \beta_2 = 0$ and $H_a: \beta_2 \neq 0$. The test statistic is:

$$t\text{-statistic} = \frac{.0040476 - 0}{.001394} = 2.90$$

with associated p-value = $\text{TDIST}(2.90, 6, 2) = 2.7\%$.

The hypothesis testing setup on whether the constant is 0 is given by $H_0: \beta_1 = 0$ and $H_a: \beta_1 \neq 0$. The test statistic is:

$$t\text{-statistic} = \frac{.6979517 - 0}{.8735} = 0.80$$

with associated p-value = $\text{TDIST}(0.80, 6, 2) = 45.5\%$.

Confidence Intervals and Hypothesis Tests on $\bar{Y}$

Suppose we run an OLS regression and find the associated regression line to be $\hat{Y} = b_1 + b_2 X$. This then says that, for $X = X_0$, our best guess of $E(Y | X = X_0)$ is $\hat{Y} = b_1 + b_2 X_0$. But, is this a good guess? How much does this guess deviate from reality? In an effort to address such questions we can calculate confidence intervals and conduct hypothesis tests on the true value of $\bar{Y}$.

Whether working with confidence intervals or hypothesis tests, we need to calculate the standard deviation of $\hat{Y}$. We will first calculate the variance and then ultimately take the square root:

$$\begin{aligned}\text{Var}(\hat{Y}) &= \text{Var}(b_1 + b_2 X_0) \\ &= \text{Var}(b_1) + \text{Var}(b_2 X_0) + 2 X_0 \text{Cov}(b_1, b_2) \\ &= \text{Var}(b_1) + X_0^2 \text{Var}(b_2) + 2 X_0 \text{Cov}(b_1, b_2)\end{aligned}$$

We have previously seen the explicit formulas for the variances and covariance (Handout #25), but they are repeated here for ease of reference:

$$\begin{aligned}\text{Var}(b_1) &= \sigma^2 \left[\frac{\sum_i X_i^2}{N \sum_i (X_i - \bar{X})^2} \right] \\ \text{Var}(b_2) &= \frac{\sigma^2}{\sum_i (X_i - \bar{X})^2} \\ \text{Cov}(b_1, b_2) &= \sigma^2 \left[\frac{-\bar{X}}{\sum_i (X_i - \bar{X})^2} \right]\end{aligned}$$

where we estimate σ^2 by $\text{MSE} = \frac{1}{N-2} \sum_i \varepsilon_i^2$.

By examining the above formulas, notice that a potentially convenient way to calculate $\text{Cov}(b_1, b_2)$ is given by:

$$\text{Cov}(b_1, b_2) = -\bar{X} \text{Var}(b_2).$$

It further makes sense that $\text{Cov}(b_1, b_2) < 0$ (in the typical case where $\bar{X} > 0$) since, as the slope of a line gets larger (i.e., rotated counterclockwise), the y-intercept gets smaller. That is, as b_2 tends to increase, b_1 tends to decrease.

- Confidence Intervals

The confidence interval for $\bar{Y}$ (given $X = X_0$) is: $\hat{Y} \pm (\#SD) \sqrt{\text{Var}(\hat{Y})}$, or equivalently $(b_1 + b_2 X_0) \pm (\#SD) \sqrt{\text{Var}(\hat{Y})}$ where $(\#SD)$ is taken from the t-table with $n - 2$ degrees of freedom.

- Hypothesis Tests

We can run hypothesis tests on the true value of $\bar{Y}$ for a given $X = X_0$. We know this true value is given by the unknown value $\beta_1 + \beta_2 X_0$, and we are estimating it as $b_1 + b_2 X_0$. The hypothesis test can be either one-tailed or two-tailed. For example, suppose we wish to test whether the true value equals $\bar{Y}_0$. The hypothesis testing setup would be:

$$H_0: \beta_1 + \beta_2 X_0 = \bar{Y}_0$$

$$H_a: \beta_1 + \beta_2 X_0 \neq \bar{Y}_0$$

Proceeding as usual in hypothesis tests, the t-statistic is then given by:

$$\text{t-statistic} = \frac{(b_1 + b_2 X_0) - \bar{Y}_0}{\sqrt{\text{Var}(\hat{Y})}}$$

This t-statistic can then be compared to the critical value of the hypothesis test.

Note that $\text{Var}(\hat{Y}) = \text{Var}(b_1) + X_0^2 \text{Var}(b_2) + 2 X_0 \text{Cov}(b_1, b_2)$ can be easily calculated if $\text{Var}(b_1)$, $\text{Var}(b_2)$ and $\text{Cov}(b_1, b_2)$ are explicitly given in an exercise. But, if not, we can plug in the expressions from the previous page to obtain:

$$\begin{aligned} \text{Var}(\hat{Y}) &= \text{Var}(b_1) + X_0^2 \text{Var}(b_2) + 2 X_0 \text{Cov}(b_1, b_2) \\ &= \sigma^2 \left[\frac{\sum_i X_i^2}{N \sum_i (X_i - \bar{X})^2} \right] + X_0^2 \sigma^2 \left[\frac{1}{\sum_i (X_i - \bar{X})^2} \right] + 2 X_0 \sigma^2 \left[\frac{-\bar{X}}{\sum_i (X_i - \bar{X})^2} \right] \end{aligned}$$

Ultimately, through algebra, this can be simplified to:

$$\text{Var}(\hat{Y}) = \sigma^2 \left(\frac{1}{N} + \frac{(X_0 - \bar{X})^2}{\sum_i (X_i - \bar{X})^2} \right)$$

which can then be used as an alternate method of calculation.

Variance-Covariance Matrix for Estimates in Stata

The confidence intervals and hypothesis tests discussed in the previous handout rely on calculating $\text{Var}(b_1)$, $\text{Var}(b_2)$ and/or $\text{Cov}(b_1, b_2)$. Performing this calculation can be laborious, especially when there are many, many data points. If you are running a regression in Stata, a matrix displaying these calculations can be summoned simply by typing ‘vce’ as a command once the regression has been run. The command ‘vce’ stands for ‘variance-covariance estimates.’

Consider the sample from Handout #30, which provides data on a student’s graduating GPA and the SAT mathematics score the student achieved upon initially entering the school. After running the linear regression of ‘GPA’ as a function of ‘SAT’ (i.e., a student’s SAT mathematics score), Stata displays the output shown on Handout #30. Then, if we subsequently type the command, ‘vce’, Stata delivers the output:

Covariance matrix of coefficients of regress model

e(V)	SAT	_cons
SAT	.000001943	
_cons	– .00120735	.76296483

The values along the diagonal provide the variances and the value off the diagonal provides the covariance. That is,

$$\text{Var}(\text{slope estimate}) = \text{Var}(b_2) = .000001943$$

$$\text{Var}(\text{y-intercept estimate}) = \text{Var}(b_1) = .76296483$$

$$\text{Cov}(\text{y-intercept estimate, slope estimate}) = \text{Cov}(b_1, b_2) = - .00120735$$

Notice this implies $\text{std_dev}(b_1) = \sqrt{.76296483} = 0.87348$ and $\text{std_dev}(b_2) = \sqrt{.000001943} = 0.00139$, which confirms the calculations under the ‘Std. Err.’ heading in the lower table of the Stata output shown on Handout #30.

Using Dummy Variables to Resolve Two-Population Hypothesis Tests

In Econ 102A we studied two-population hypothesis tests. Specifically, one may want to test whether the mean value of some statistic of interest is the same for two separate populations. The general procedure for running the hypothesis test is:

- (1) Set up null and alternative hypotheses (typically, $H_0: \mu_1 - \mu_2 = 0$ and $H_a: \mu_1 - \mu_2 \neq 0$).
- (2) Take a sample from both populations and calculate the sample means ($\bar{x}_1$ and $\bar{x}_2$) and sample standard deviations (s_1 and s_2) respectively.
- (3) Calculate the t-statistic by standardizing the value $\bar{x}_1 - \bar{x}_2$ and seeing how many standard deviations it strays from zero.
- (4) Reach a conclusion on the null hypothesis based on comparing the t-statistic to the critical value (or, equivalently, comparing the p-value to the level of significance).

As an alternative to performing the above steps, the hypothesis test can be equivalently performed using regression. Instead of thinking about the sample data as coming from two, separate groups we instead combine the data into a single sample. We will distinguish which data points come from which population by assigning all data points within one population $X = 0$ and all data points from the other population as $X = 1$; this is the introduction of the dummy variable into the regression. The specific values of the variable gleaned in the original samples will serve as the Y-variable.

For example, suppose we wish to test whether employees in Company A make, on average, a higher salary than employees in Company B. We sample three employees from Company A who have salaries of \$55,000, \$109,000 and \$73,000; we sample two employees from Company B who have salaries of \$64,000 and \$89,000. Our data set may then be represented as:

Company (X)	Salary (Y)
0	\$55,000
0	\$109,000
0	\$73,000
1	\$64,000
1	\$89,000

where we have arbitrarily assigned $X = 0$ to Company A employees and $X = 1$ to Company B employees.

We can then perform the two-population hypothesis test (on whether the two companies have a statistically significant difference in average salaries) by running a regression of Y on X .

The details of this procedure will be done via an example in lecture. This handout presents some of the mathematical results that will be alluded to during the presentation.

- Calculating the y-intercept and slope of the regression line

The population regression line is $\bar{Y} = \beta_1 + \beta_2 X$. So, when $X = 0$, we have $E(Y | X = 0) = \beta_1$. Our best guess of β_1 (i.e., b_1) will therefore be the average y-value for all $X = 0$ observations in the sample. Further, when $X = 1$, we have $E(Y | X = 1) = \beta_1 + \beta_2$. So, $\beta_2 = E(Y | X = 1) - \beta_1 = E(Y | X = 1) - E(Y | X = 0)$. Therefore, our best guess of β_2 (i.e., b_2) will be the difference in the average y-value for all $X = 1$ observations in the sample minus the average y-value for all $X = 0$ observations in the sample.

$\Rightarrow b_1 =$ the sample mean of y-values for the observations assigned $X = 0$

$b_2 =$ (the sample mean of y-values for the observations assigned $X = 1$) – (the sample mean of y-values for the observations assigned $X = 0$)

- The variance of the Y-variable for a given X-value

Recall that an assumption of simple linear regression is that $\text{Var}(Y | X = X_0)$ is constant, regardless of the value of X_0 . In other words, we assume our two populations have equal variances. When making this assumption in two-population hypothesis tests we estimate the (common) population variance as:

$$s_{pooled}^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}$$

In a previous handout we have seen that, in a regression setting, $\text{Var}(Y | X = X_0) =$

$\text{Var}(\varepsilon | X = X_0)$ can be estimated as $\text{MSE} = \frac{1}{N - 2} \sum_i \varepsilon_i^2$.

$\Rightarrow$ the common pooled variance we saw in the realm of two-population hypothesis testing is the same as the mean sum of squared residuals in the realm of regression. That is, for a simple regression which uses a dummy variable as a predictor, we have:

$$s_{pooled}^2 = \text{MSE}$$

- The variance of b_1

A previous bullet point has shown that $b_1 =$ the sample mean of y-values for the observations assigned $X = 0$. Therefore, $\text{Var}(b_1) =$ the variance of the sample mean of y-values for the

observations assigned $X = 0$. In Econ 102A we saw that, in general, the variance for a sample mean is $\frac{\sigma^2}{n}$ where σ^2 is the population variance of the variable, and n is the number of observations being averaged. Since we estimate the population variance of y -values for the observations assigned $X = 0$ as $s_{pooled}^2 = \text{MSE}$, we therefore have:

For a simple regression which uses a dummy variable as a predictor:

$$\Rightarrow \text{Var}(b_1) = \frac{\sigma^2}{n} = \frac{\text{MSE}}{n_{x=0}} = \frac{s_{pooled}^2}{n_{x=0}}$$

where, in the above expression, $n_{x=0}$ is the number of sample observations assigned $X = 0$.

- The variance of b_2

A previous bullet point has shown that $b_2 =$ the difference in sample means of y -values. Specifically, $b_2 =$ (the sample mean of y -values for the observations assigned $X = 1$) – (the sample mean of y -values for the observations assigned $X = 0$). Therefore, $\text{Var}(b_2)$ is the variance of this difference in sample means, and we consequently have:

$$\begin{aligned} \text{Var}(b_2) &= \text{Var}(\bar{y}_{x=1} - \bar{y}_{x=0}) \\ &= \text{Var}(\bar{y}_{x=1}) + \text{Var}(-\bar{y}_{x=0}) \quad (\text{by the independence of the samples}) \\ &= \text{Var}(\bar{y}_{x=1}) + (-1)^2 \text{Var}(\bar{y}_{x=0}) \quad (\text{by properties of variance}) \\ &= \text{Var}(\bar{y}_{x=1}) + \text{Var}(\bar{y}_{x=0}) \end{aligned}$$

As noted above, in Econ 102A, we saw that the variance for a sample mean is $\frac{\sigma^2}{n}$ where σ^2 is the population variance of the variable, and n is the number of observations being averaged.

Since we estimate the population variance of y -values (conditional on either $X = 0$ or $X = 1$) as $s_{pooled}^2 = \text{MSE}$, we therefore have:

$$\begin{aligned} \text{Var}(b_2) &= \text{Var}(\bar{y}_{x=1}) + \text{Var}(\bar{y}_{x=0}) \\ &= \frac{\text{MSE}}{n_{x=1}} + \frac{\text{MSE}}{n_{x=0}} \\ &= \frac{s_{pooled}^2}{n_{x=1}} + \frac{s_{pooled}^2}{n_{x=0}} \end{aligned}$$

So, for a simple regression which uses a dummy variable as a predictor:

$$\Rightarrow \text{Var}(b_2) = \frac{\text{MSE}}{n_{x=1}} + \frac{\text{MSE}}{n_{x=0}} = \frac{s_{\text{pooled}}^2}{n_{x=1}} + \frac{s_{\text{pooled}}^2}{n_{x=0}}$$

where, in the above expression, $n_{x=1}$ is the number of sample observations assigned $X = 1$ and $n_{x=0}$ is the number of sample observations assigned $X = 0$.

- Running a two-population hypothesis test on the difference in means by using a dummy-variable, simple regression

In Econ 102A, when we conduct a two-population hypothesis test on means, we standardize the ‘difference in sample means’ by dividing by its standard deviation. Let us now reinforce how this same hypothesis test can be done through a dummy-variable, simple regression.

The variance calculation required to ultimately standardize the ‘difference in sample means’ is given by $\text{Var}(\bar{y}_{x=1}) + \text{Var}(\bar{y}_{x=0})$. And, we have already established $\bar{y}_{x=0} = b_1$ and $\bar{y}_{x=1} = b_1 + b_2$. So, in terms of b_1 and b_2 , we have:

$$\begin{aligned} \text{Var}(\bar{y}_{x=1}) + \text{Var}(\bar{y}_{x=0}) &= \text{Var}(b_1 + b_2) + \text{Var}(b_1) \\ &= \text{Var}(b_1) + \text{Var}(b_2) + 2 \text{Cov}(b_1, b_2) + \text{Var}(b_1) \\ &= 2 \text{Var}(b_1) + \text{Var}(b_2) + 2 \text{Cov}(b_1, b_2) \quad (**) \end{aligned}$$

From a previous handout we have seen:

$$\text{Var}(b_1) = \sigma^2 \left[\frac{\sum_i X_i^2}{N \sum_i (X_i - \bar{X})^2} \right] \quad \text{and} \quad \text{Cov}(b_1, b_2) = \sigma^2 \left[\frac{-\bar{X}}{\sum_i (X_i - \bar{X})^2} \right].$$

When X is a dummy variable, $\sum_i X_i^2 =$ the number of observations in the sample assigned

$X = 1$. Therefore, $\sum_i X_i^2 / N =$ the *proportion* of observations in the sample assigned $X = 1$.

In other words, $\sum_i X_i^2 / N = \bar{X}$. So,

$$\text{Var}(b_1) + \text{Cov}(b_1, b_2) = \sigma^2 \left[\frac{\bar{X}}{\sum_i (X_i - \bar{X})^2} \right] + \sigma^2 \left[\frac{-\bar{X}}{\sum_i (X_i - \bar{X})^2} \right] = 0.$$

Now, going back to (**), we therefore have:

$$2 \text{Var}(b_1) + \text{Var}(b_2) + 2 \text{Cov}(b_1, b_2) = 2 [\text{Var}(b_1) + \text{Cov}(b_1, b_2)] + \text{Var}(b_2)$$

which, using the result we just developed, leads to:

$$2 [\text{Var}(b_1) + \text{Cov}(b_1, b_2)] + \text{Var}(b_2) = 2 (0) + \text{Var}(b_2) = \text{Var}(b_2).$$

Overall, this derivation simply confirms that the variance (and hence standard deviation) of the ‘difference in means’ used in Econ 102A to run our two-population hypothesis tests on means equals the variance (and hence standard deviation) of b_2 in a simple regression which uses a dummy variable as a predictor.

In summary, we now realize how to run a two-population hypothesis test on the ‘difference in means’ by appealing to the Stata output for a simple regression which uses a dummy variable as a predictor. Specifically, when we knew nothing of regression in Econ 102A we calculated the t-statistic of the hypothesis test (for the null hypothesis $H_0: \mu_1 - \mu_2 = 0$) as:

$$\text{t-statistic} = \frac{\text{difference in sample means}}{\text{std dev}(\text{difference in sample means})}.$$

Now, by the discussion of this handout, the null hypothesis translates as $H_0: \beta_2 = 0$ and this same test statistic translates as:

$$\text{t-statistic} = \frac{b_2}{\text{std dev}(b_2)}.$$

Recall that this test is already run naturally as part of the Stata output. So, to test whether there is a significant difference in sample means between two populations, we can run a regression which uses a dummy-variable as the predictor and then simply look at the p-value associated with the dummy variable in the lower table of the Stata output.

Prediction Intervals

In class we will develop the concept of a ‘prediction interval.’ The concept is related to confidence intervals but the distinction between the two is important. Specifically, a confidence interval gives a range of values hoping to capture *the overall population average* of a statistic of interest, whereas a prediction interval gives a range of values hoping to capture *one specific observation* from the population of interest. For example, we might consider the selling price of all homes in Sacramento, California. A confidence interval would attempt to capture the average selling price of all the thousands of homes in Sacramento, California. A prediction interval, by contrast, would attempt to capture the selling price of just the single home located at 500 Main Street in Sacramento, California.

As we will develop in class, the prediction interval relies on determining the standard deviation of the quantity $(X_{\text{new}} - \bar{X})$, where $\bar{X}$ is the mean of the sample taken and X_{new} is any new observation from the population. It should be noted that, for the prediction intervals we do in class, we assume the population is normally distributed. We will also assume here that the population distribution has variance σ^2 .

In terms of determining the standard deviation of interest, we have:

$$\begin{aligned}
 \text{Var}(X_{\text{new}} - \bar{X}) &= \text{Var}(X_{\text{new}}) + \text{Var}(-\bar{X}) \quad (\text{because of independence}) \\
 &= \text{Var}(X_{\text{new}}) + (-1)^2 \text{Var}(\bar{X}) \quad (\text{due to properties of variance}) \\
 &= \text{Var}(X_{\text{new}}) + \text{Var}(\bar{X}) \\
 &= \sigma^2 + \frac{\sigma^2}{N} \quad (\text{the first term is by assumption of the population} \\
 &\quad \text{variance; the second term is due to the CLT of Means} \\
 &\quad \text{from Econ 102A}) \\
 &= \sigma^2 \left(1 + \frac{1}{N} \right)
 \end{aligned}$$

Therefore, the standard deviation of $(X_{\text{new}} - \bar{X}) = \sqrt{\sigma^2 \left(1 + \frac{1}{N} \right)} = \sigma \sqrt{1 + \frac{1}{N}}$.

Since the population variance will typically be unknown, we estimate it by using the sample variance, s^2 . Hence, the standard deviation of $(X_{\text{new}} - \bar{X})$ is taken to be $s \sqrt{1 + \frac{1}{N}}$.

Prediction Intervals in Regression

In the previous handout on prediction intervals, there is only one variable of interest. We have supposedly taken a sample of size N from a population and, based on that sample, we wish to determine an interval that predicts the value of the $(N+1)^{\text{st}}$ observation which we are about to take from that same (normal) population. In this situation, the previous handout shows that the overall variation is composed of two parts: $\text{Var}(X_{\text{new}})$ and $\text{Var}(\bar{X})$. That is, there is variation coming from (1) the variance in the population from which the observation is drawn (this is measured through $\text{Var}(X_{\text{new}})$), and (2) the variation in the sample mean possibly deviating from the population mean (this is measured through $\text{Var}(\bar{X})$).

When applying the concept of prediction intervals to regression the logic is analogous. Here, though, we are trying to predict a specific Y -value based on a given X -value. That is, we are no longer predicting a single X -value within the population. Instead, we are predicting a single Y -value within the population given a (known) X -value. Luckily, the process is the same: the overall variation breaks down into the exact same two variation components cited in the previous handout (namely, $\text{Var}(Y_{\text{new}})$ and $\text{Var}(\hat{Y})$). Performing the analogous calculation for the Y -values that we saw in the last handout for the X -values, we have:

$$\begin{aligned}\text{Var}(Y_{\text{new}} - \hat{Y}) &= \text{Var}(Y_{\text{new}}) + \text{Var}(-\hat{Y}) \quad (\text{because of independence}) \\ &= \text{Var}(Y_{\text{new}}) + (-1)^2 \text{Var}(\hat{Y}) \quad (\text{due to properties of variance}) \\ &= \text{Var}(Y_{\text{new}}) + \text{Var}(\hat{Y}).\end{aligned}$$

$\text{Var}(Y_{\text{new}})$ is just the variance of the population distribution of Y (for a given X -value), which we know to be estimated by MSE. And, we have previously seen that:

$$\text{Var}(\hat{Y}) = \text{Var}(b_1) + X_0^2 \text{Var}(b_2) + 2 X_0 \text{Cov}(b_1, b_2)$$

Therefore, the proper variance to use for a prediction interval in a regression setting (i.e., when trying to predict a single observation of $(Y | X = X_0)$ from the population) is:

$$\text{Var}(Y_{\text{new}}) + \text{Var}(\hat{Y}) = \text{MSE} + \text{Var}(b_1) + X_0^2 \text{Var}(b_2) + 2 X_0 \text{Cov}(b_1, b_2)$$

As an alternate approach, we have previously seen that:

$$\text{Var}(\hat{Y}) = \sigma^2 \left(\frac{1}{N} + \frac{(X_0 - \bar{X})^2}{\sum_i (X_i - \bar{X})^2} \right)$$

so, $\text{Var}(Y_{\text{new}}) + \text{Var}(\hat{Y})$ can be expressed as:

$$\begin{aligned}
&= \sigma^2 + \sigma^2 \left(\frac{1}{N} + \frac{(X_0 - \bar{X})^2}{\sum_i (X_i - \bar{X})^2} \right) \\
&= \sigma^2 \left(1 + \frac{1}{N} + \frac{(X_0 - \bar{X})^2}{\sum_i (X_i - \bar{X})^2} \right)
\end{aligned}$$

where we estimate σ^2 by $\text{MSE} = \frac{1}{N-2} \sum_i \varepsilon_i^2$.

As a side note, recall that $\text{Var}(b_2) = \frac{\sigma^2}{\sum_i (X_i - \bar{X})^2}$. Therefore, our variance calculation can

also be computed as:

$$\begin{aligned}
\sigma^2 \left(1 + \frac{1}{N} + \frac{(X_0 - \bar{X})^2}{\sum_i (X_i - \bar{X})^2} \right) &= \sigma^2 + \frac{\sigma^2}{N} + \frac{\sigma^2 (X_0 - \bar{X})^2}{\sum_i (X_i - \bar{X})^2} \\
&= \sigma^2 + \frac{\sigma^2}{N} + (X_0 - \bar{X})^2 \frac{\sigma^2}{\sum_i (X_i - \bar{X})^2} \\
&= \sigma^2 \left(1 + \frac{1}{N} \right) + (X_0 - \bar{X})^2 \text{Var}(b_2).
\end{aligned}$$

Since the natural Stata output essentially provides σ^2 (through MSE) and $\text{Var}(b_2)$, this latter expression may provide a quick way to calculate the prediction interval variance.

Overall, then, the variance used for generating a prediction interval on a single Y-value from the population (for a given X-value) can be computed by any of the following means:

- $\text{MSE} + \text{Var}(b_1) + X_0^2 \text{Var}(b_2) + 2 X_0 \text{Cov}(b_1, b_2)$

- $\text{MSE} \left(1 + \frac{1}{N} + \frac{(X_0 - \bar{X})^2}{\sum_i (X_i - \bar{X})^2} \right)$

- $\text{MSE} \left(1 + \frac{1}{N} \right) + (X_0 - \bar{X})^2 \text{Var}(b_2)$

R²-value (or Coefficient of Determination)

We have shown how to use a sample of data to determine a regression line, and much of our previous work has been devoted to finding values for b_1 , b_2 and $\hat{Y}$ (along with their associated variances) and using these values to construct confidence intervals and run hypothesis tests on population parameters.

However, we have not yet addressed a rather fundamental question: how well does a given regression line fit its original scatter diagram? Our initial process was simply to construct a scatter diagram of the sample observations, reflect on the diagram, hope the trend looks fairly linear and, if so, forge ahead with an OLS regression line. But is there some way we can quantify how well the regression line fits the data? After all, one of the fundamental uses of regression is to hopefully make an accurate prediction of Y based on a given X -value. If the sample data we have gleaned aligns itself in linear fashion, this bodes well that a linear model will give us accurate predictions in the future. Ultimately, one way to quantify how well a linear model fits the underlying data is through the R^2 -value, also known as the coefficient of determination.

An explanation of this concept will be given in lecture. Here, though, we present the underlying mathematics in an effort to facilitate the classroom discussion. The following development is different than that presented in lecture to add variety (although the eventual conclusion is the same):

Overall, once a regression line is determined, we have $Y = \hat{Y} + \varepsilon$. Taking the variance of both sides gives:

$$\text{Var}(Y) = \text{Var}(\hat{Y}) + \text{Var}(\varepsilon) + 2 \text{Cov}(\hat{Y}, \varepsilon)$$

but, because error terms are independent of predicted values, $\text{Cov}(\hat{Y}, \varepsilon) = 0$, so we have:

$$\text{Var}(Y) = \text{Var}(\hat{Y}) + \text{Var}(\varepsilon)$$

$$\Rightarrow E[(Y - \bar{Y})^2] = E[(\hat{Y} - \bar{Y})^2] + E[(\varepsilon - \bar{\varepsilon})^2].$$

Recalling that an assumption of OLS regression is that $\bar{\varepsilon} = 0$, this leads to:

$$\Rightarrow E[(Y - \bar{Y})^2] = E[(\hat{Y} - \bar{Y})^2] + E[(\varepsilon)^2]$$

$$\Rightarrow \frac{\sum_i (Y_i - \bar{Y})^2}{N} = \frac{\sum_i (\hat{Y}_i - \bar{Y})^2}{N} + \frac{\sum_i \varepsilon_i^2}{N}$$

Multiplying through by N gives:

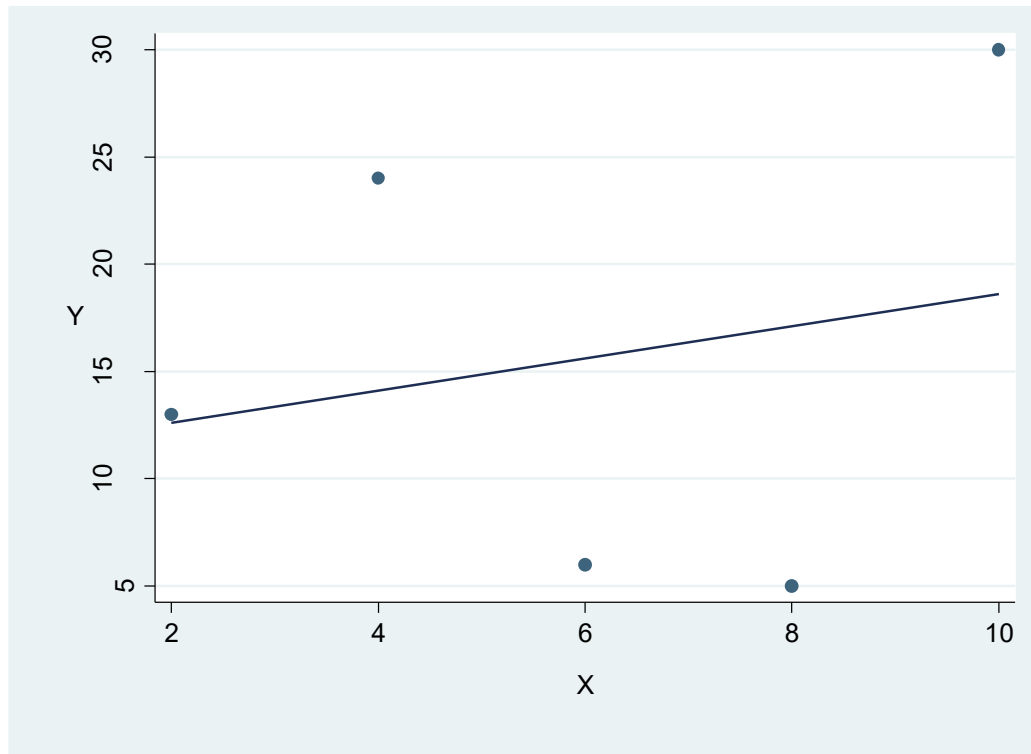
$$\Rightarrow \sum_i (Y_i - \bar{Y})^2 = \sum_i (\hat{Y}_i - \bar{Y})^2 + \sum_i \varepsilon_i^2$$

$$\Rightarrow \text{SST} = \text{SSR} + \text{SSE}$$

This is a general relationship which holds true in all regressions. This result will be cited in lecture and is further confirmed through the 'Elaboration of Worksheet Question 5' on Page 7 of 9 of the Week 4 through Week 6 Worksheet (Handout #28).

Visualizing the R²-value

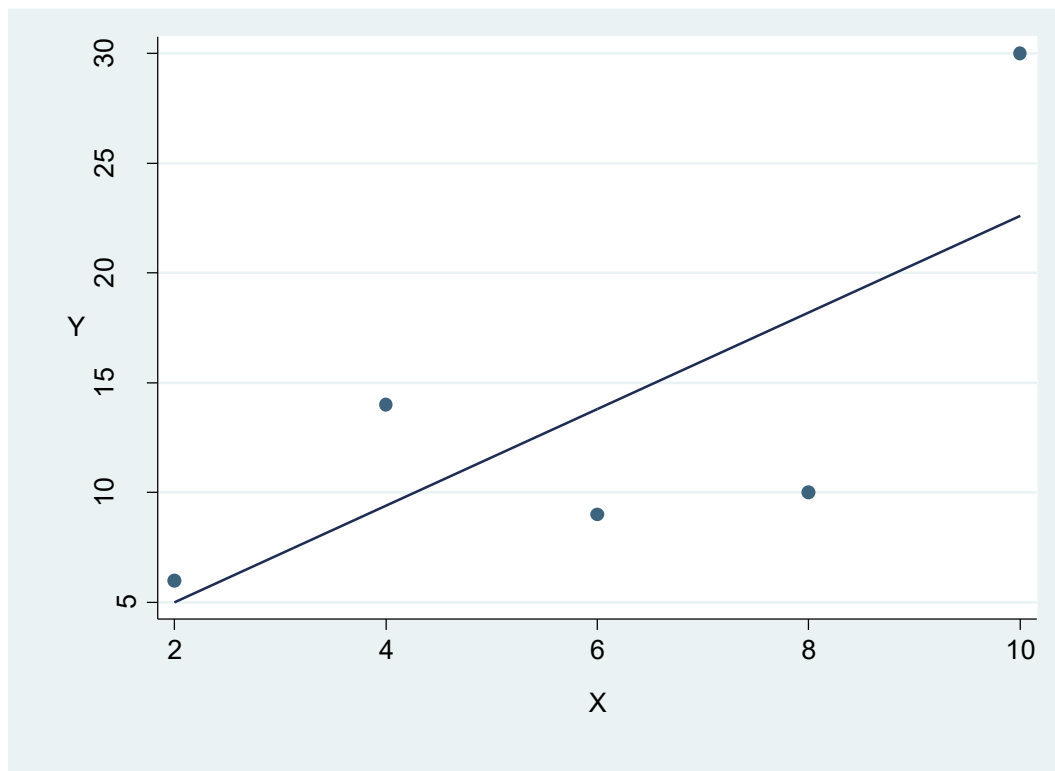
The following graphs show various sample observations and their associated regression line. Notice that the more the sample observations conform to a linear pattern, the higher the consequent R²-value.



Regression Line: $\hat{Y} = 0.75 X + 11.1$

X	Y	$\hat{Y}$	$(Y - \bar{Y})$	$(Y - \bar{Y})^2$	$(\hat{Y} - \bar{Y})$	$(\hat{Y} - \bar{Y})^2$
2	13	12.6	-2.6	6.76	-3.0	9.00
4	24	14.1	8.4	70.56	-1.5	2.25
6	6	15.6	-9.6	92.16	0	0
8	5	17.1	-10.6	112.36	1.5	2.25
10	30	18.6	14.4	207.36	3.0	9.00
SST				489.20	SSR	22.50

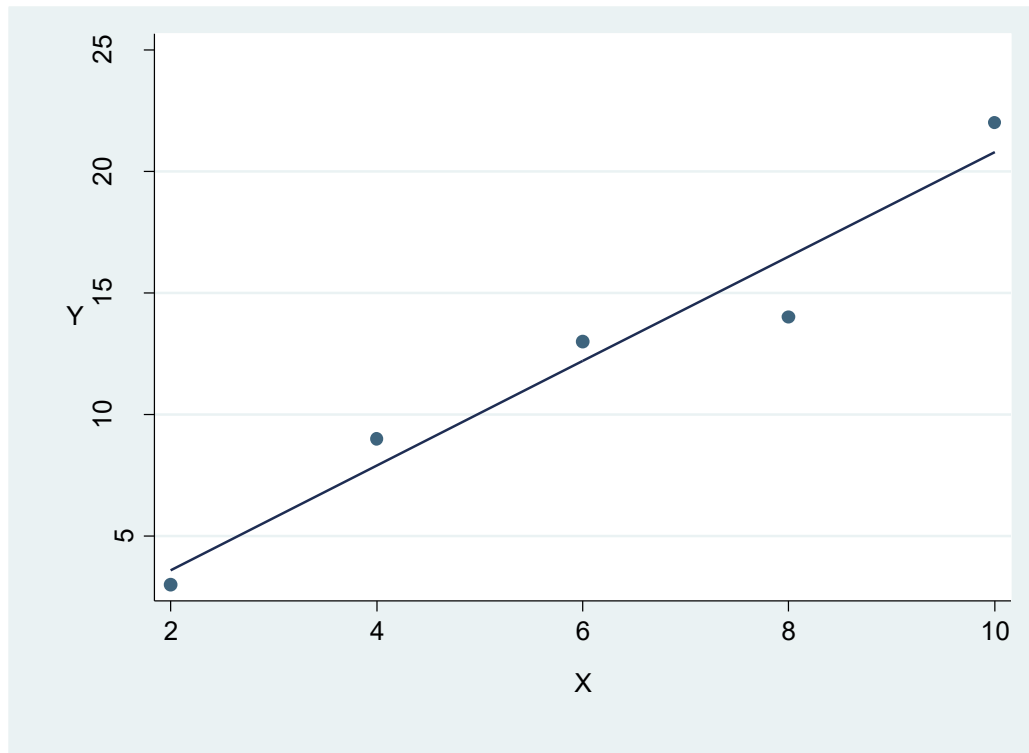
$$\Rightarrow R^2\text{-value} = \frac{SSR}{SST} = \frac{22.50}{489.20} = 4.60\%$$



Regression Line: $\hat{Y} = 2.2 X + 0.6$

X	Y	$\hat{Y}$	$(Y - \bar{Y})$	$(Y - \bar{Y})^2$	$(\hat{Y} - \bar{Y})$	$(\hat{Y} - \bar{Y})^2$
2	6	5.0	-7.8	60.84	-8.8	77.44
4	14	9.4	0.2	0.04	-4.4	19.36
6	9	13.8	-4.8	23.04	0	0
8	10	18.2	-3.8	14.44	4.4	19.36
10	30	22.6	16.2	262.44	8.8	77.44
SST				360.80	SSR	193.60

$$\Rightarrow R^2\text{-value} = \frac{SSR}{SST} = \frac{193.60}{360.80} = 53.66\%$$



Regression Line: $\hat{Y} = 2.15 X - 0.7$

X	Y	$\hat{Y}$	$(Y - \bar{Y})$	$(Y - \bar{Y})^2$	$(\hat{Y} - \bar{Y})$	$(\hat{Y} - \bar{Y})^2$
2	3	3.6	-9.2	84.64	-8.6	73.96
4	9	7.9	-3.2	10.24	-4.3	18.49
6	13	12.2	0.8	0.64	0	0
8	14	16.5	1.8	3.24	4.3	18.49
10	22	20.8	9.8	96.04	8.6	73.96
SST				194.80	SSR	184.90

$$\Rightarrow R^2\text{-value} = \frac{SSR}{SST} = \frac{184.90}{194.80} = 94.92\%$$

Another Look at Stata Output

Recall the example we saw earlier regarding a University Admissions Office running a regression on student's graduating GPA against student's incoming SAT mathematics score. As before, the data for eight of the school's students are:

SAT Math Score (X)	Graduating GPA (Y)
640	3.4
700	3.2
550	3.5
540	2.8
620	3.0
490	2.4
710	3.6
720	3.8

We used this example to explicitly show how Stata output relates to some of our class calculations to date. There were some items from the Stata output which we did not comment on, but now we can. Recall the Stata output for this example:

Source	SS	df	MS	
Model	.869719965	1	.869719965	
Residual	.619029824	6	.103171637	
Total	1.48874979	7	.212678541	

Number of obs = 8
 F(1, 6) = 8.43
 Prob > F = 0.0272
 R-squared = 0.5842
 Adj R-squared = 0.5149
 Root MSE = .3212

GPA	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
SAT	.0040476	.0013941	2.90	0.027	.0006364	.0074587
_cons	.6979517	.8734786	0.80	0.455	-1.439373	2.835277

Elaboration on the above calculations:

1. The sum of squares for the regression line (which, in our class discussion is labeled SSR and, in Stata, corresponds to the table entry in row 'Model' and column 'SS') is based on the following table:

X	Y	$\hat{Y}$
640	3.4	3.288
700	3.2	3.531
550	3.5	2.924
540	2.8	2.884
620	3.0	3.207
490	2.4	2.681
710	3.6	3.572
720	3.8	3.612

Recalling that the mean of Y is given by:

$$E(Y) = \frac{1}{8}(3.4 + 3.2 + 3.5 + 2.8 + 3.0 + 2.4 + 3.6 + 3.8) = 3.2125$$

we have:

$$\begin{aligned} SSR &= \sum_i (\hat{Y}_i - \bar{Y})^2 = (3.288 - 3.2125)^2 + (3.531 - 3.2125)^2 + (2.924 - 3.2125)^2 + \\ &\quad (2.884 - 3.2125)^2 + (3.207 - 3.2125)^2 + (2.681 - 3.2125)^2 + \\ &\quad (3.572 - 3.2125)^2 + (3.612 - 3.2125)^2 \\ &= .00570 + .10144 + .08323 + .10791 + .00003 + .28249 + .12924 + .15960 = .8697. \end{aligned}$$

2. In general, the total degrees of freedom for the entire sample of data is $N - 1$, whereas the specific degrees of freedom for the residuals is $N - K$, where K is the number of parameters being estimated by the regression line. So, taking the difference, the degrees of freedom for the model (which is also equal to the number of X-variables used as predictors) is therefore $(N - 1) - (N - K) = K - 1$.

Since $K = 2$ here (we are estimating both β_1 and β_2) the degrees of freedom for the model is therefore $2 - 1 = 1$.

Thus, the mean sum of squares for the regression line is $\frac{SSR}{df} = \frac{.8697}{1} = .8697$.

3. The total sum of squares (which, in our class discussion is labeled SST and, in Stata, corresponds to the table entry in row 'Total' and column 'SS') is the sum of squares for the regression line added to the sum of squared residuals. That is, $SST = SSR + SSE$.

The sum of squared residuals has been calculated on a previous 'Stata output' handout (Hanout #30) as:

$$\sum_i \varepsilon_i^2 = .0125 + .1097 + .3316 + .0070 + .0430 + .0791 + .0008 + .0353 = .6190.$$

Further, we calculated SSR above as .8697. Therefore, $SST = .8697 + .6190 = 1.4887$.

4. In general, the total degrees of freedom for the entire sample of data is $N - 1$. Since the sample size is $n = 8$, the total degrees of freedom is $8 - 1 = 7$.

Thus, the mean total sum of squares is $\frac{SST}{total\ df} = \frac{1.4887}{7} = .2127$.

5. The R^2 -value is given by $\frac{SSR}{SST} = \frac{.8697}{1.4887} = .5842$.

Common Nonlinear Functional Forms

When considering the logical relationship between regression variables, or when directly viewing a scatter diagram of the data, one might surmise the relationship between X and Y to be nonlinear. Indeed, there are a variety of common nonlinear functional forms within the modeler's arsenal. This handout outlines these common functional forms.

- Common Nonlinear Functional Forms:

(1) Linear-Log Model: $Y = \beta_1 + \beta_2 \ln(X)$

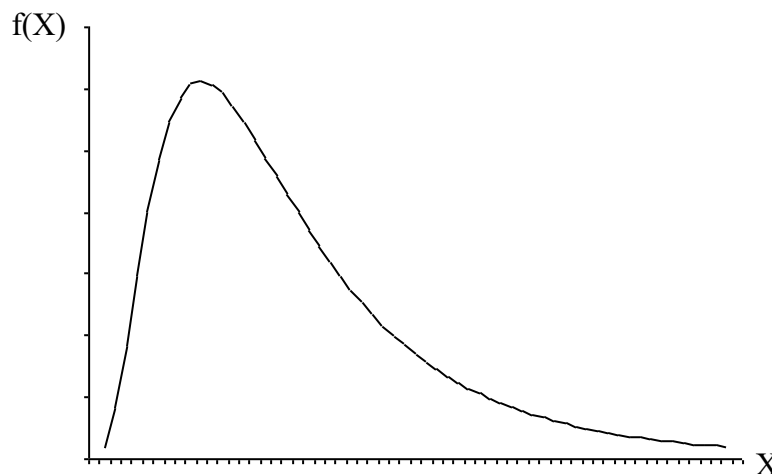
(2) Log-Linear Model: $\ln(Y) = \beta_1 + \beta_2 X$

(3) Log-Log Model: $\ln(Y) = \beta_1 + \beta_2 \ln(X)$

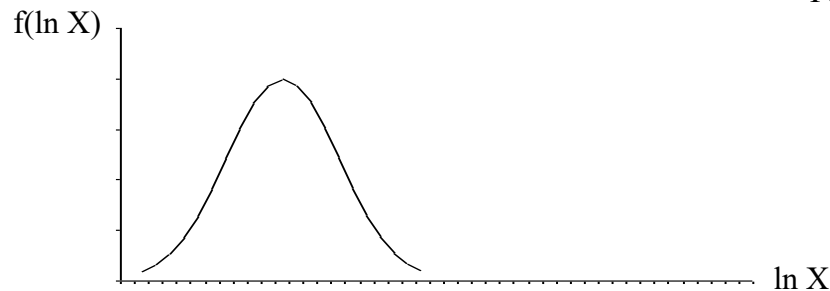
(4) Quadratic Model: $Y = \beta_1 + \beta_2 X^2$

(5) Cubic Model: $Y = \beta_1 + \beta_2 X^3$

You will notice, in particular, that several of the above functional forms involve the logarithm of one or more of the variables. What is the impetus for using a logarithmic transformation on a variable? Many variables (such as annual income earned, housing prices and/or investment balances) have a distribution truncated on the left at zero, concentrated heavily within a range, but having a long tail to the right (i.e., there is the possibility for very high observations of the variable, albeit with a fairly low probability). Visually such variables, X , look like:



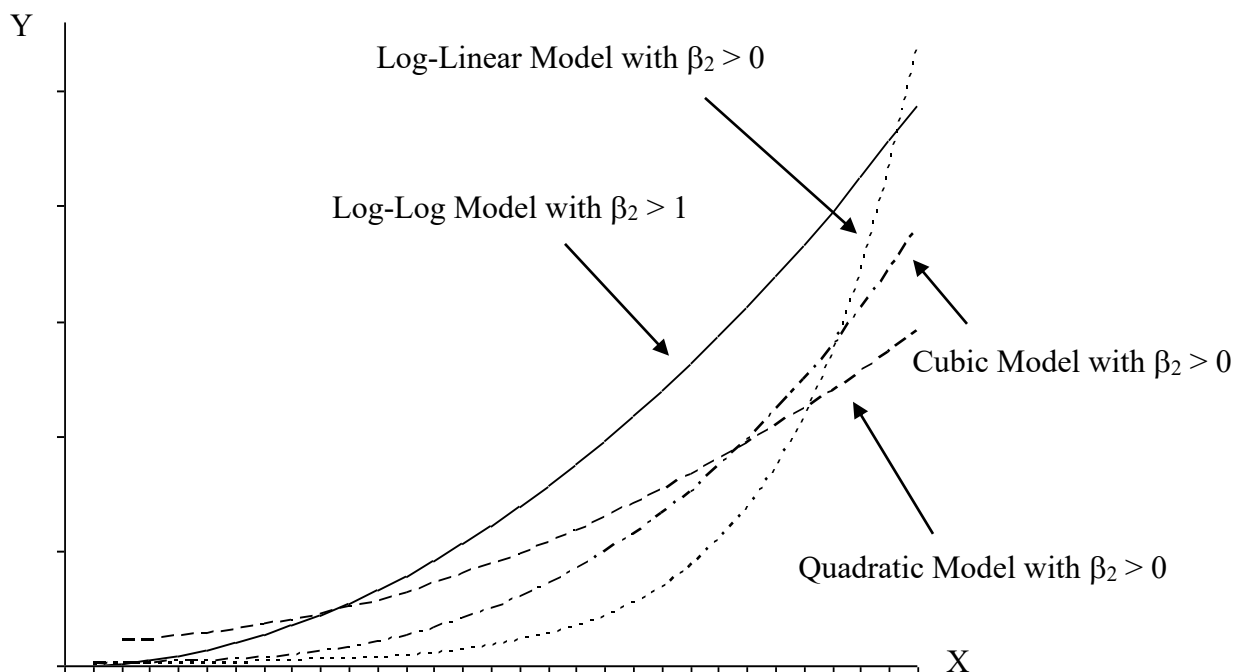
But, when one measures the *logarithm of the variable* on the X -axis instead of the original variable, the same distribution from above becomes:



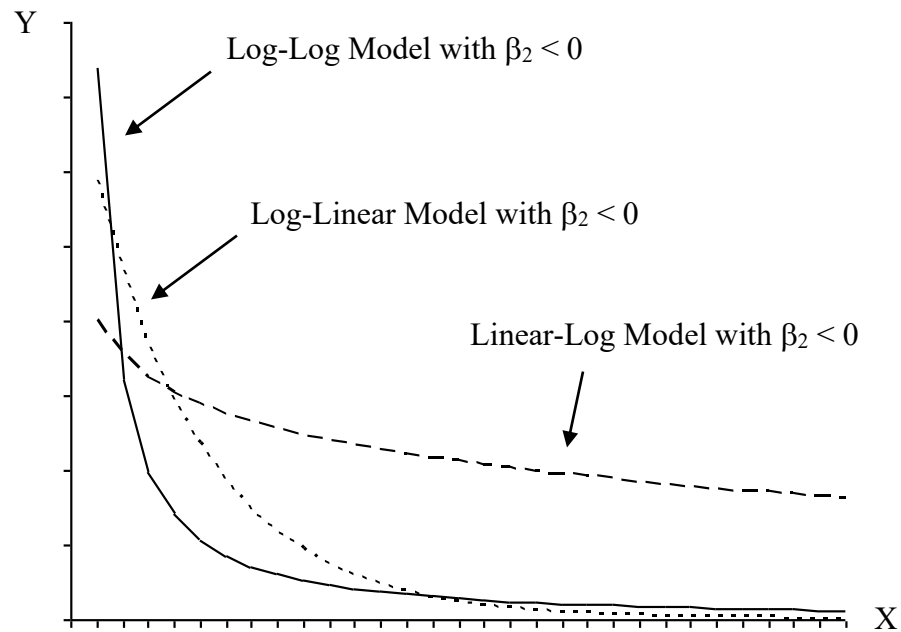
and we see that it mimics a normal distribution. In this way the original distribution of X is referred to as a *log-normal distribution*. The consequence is that running a regression on Y against $\ln(X)$ can often result in a model with better predictive ability and greater adherence to the linear regression assumptions than does running a model on Y against the original X . Specifically, if the sample contains many observations with high X -values, these X -values serve as outliers when regressing Y against X (as these X -values stray quite far from the majority of the X -distribution). These outliers can heavily influence the regression line, thus affecting the predictive ability for the (highly-concentrated) mainstream X -values in the distribution. But, when regressing Y against $\ln(X)$, these same observations in your sample are not as deviant, and therefore have a lesser effect on the overall ensuing regression line. This may well lead to better predictions for the X -values in the midrange of the distribution (which constitute a high proportion of the population). And, having fewer outliers in the original sample enhances the chance that the homoskedasticity assumption is not violated.

So, from a visual standpoint, what do each of the nonlinear functional forms look like? Below, you will find several categories and the functional forms which adhere to those categories:

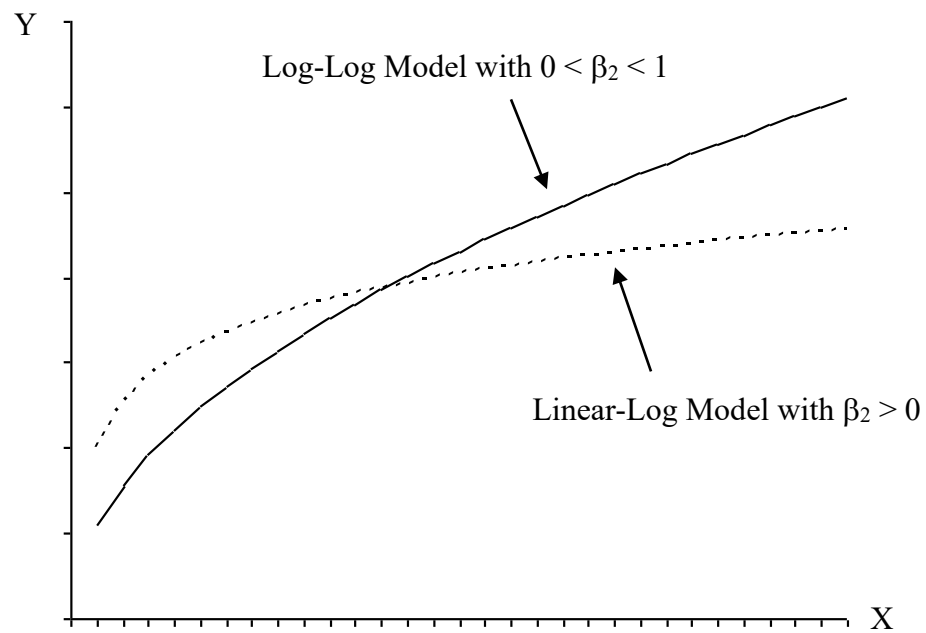
- Convex Increasing Functions:



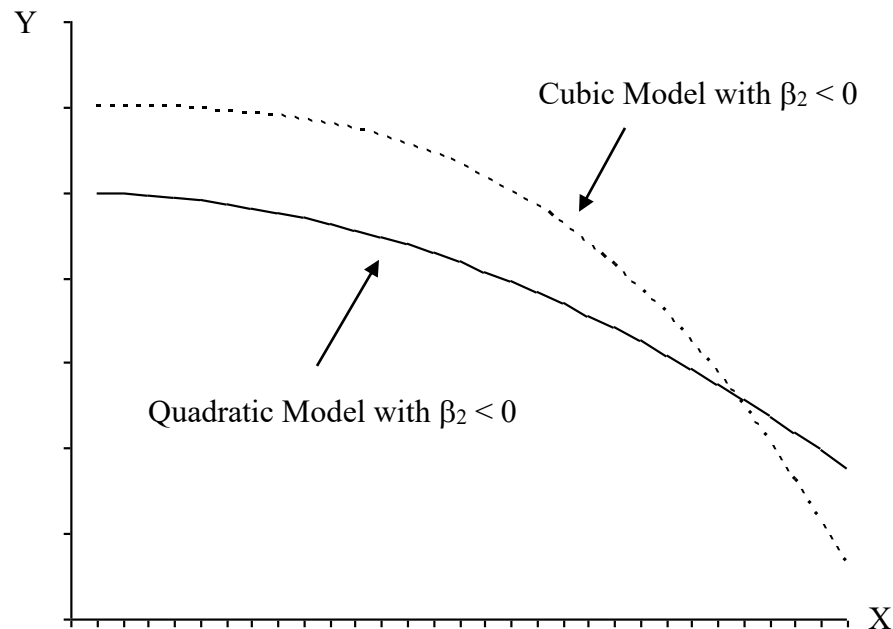
- Convex Decreasing Functions:



- Concave Increasing Functions:



- Concave Decreasing Functions:



It is also important to interpret the slope coefficient, β_2 , for the logarithmic models. Specifically, we have:

- For a log-log model β_2 represents the (approximate) percent change in Y for a 1% change in X. That is, β_2 represents the *elasticity* of the model. Since β_2 is constant, log-log models are referred to as ‘constant elasticity’ models.
- For a log-linear model β_2 embeds the percent change in Y for a 1-unit change in X. Specifically, for a 1-unit change in X, Y increases by approximately $(1 + \beta_2)\%$. Since Y increases by a constant percent whenever X increases by 1 unit, log-linear models are sometimes referred to as ‘constant growth’ models.
- For a linear-log model β_2 embeds the (approximate) unit change in Y for a 1% change in X. Specifically, for a 1% change in X, Y increases by approximately $\beta_2/100$ units. Linear-log models are useful when Y increases as X increases, but at a diminishing rate (e.g., each dollar a company spends on advertising (the X-variable) likely has a positive yet diminishing effect on the amount of company sales (the Y-variable)).

Overall, there is no objective way to determine the ‘proper’ functional form to use for a given data set. However, a couple guidelines would include:

- (1) The functional form should be consistent with economic theory (what sign do you anticipate for the first derivative? The second derivative?) and should be consistent with any economic assumptions governing the exercise (e.g., constant elasticity or constant growth over time). (Please see Week 4 through Week 6 Worksheet Questions 6 and 7 respectively.)

- (2) One might define the ‘best’ fitting model as the one with the highest R^2 -value, previously defined as $\frac{SSR}{SST}$. However, it should be noted that a comparison of such R^2 -values across different models is only valid if:
- (i) both models have the same dependent variable (i.e., we cannot compare a model using Y on the left-hand side versus one using $\ln(Y)$), and
 - (ii) there is only one, single X -variable as a predictor. (A general fact about R^2 -values is that, whenever a model increases the number of X -variables as predictors, the R^2 -value of the ensuing regression line will necessarily increase. In this way, one can attain virtually any R^2 -value one desires for a regression line simply by adding more and more X -variables as predictors).
- (3) If the assumptions stated in guideline (2) above are not satisfied, we might look at the correlation between the actual Y -values in the sample versus the Y -values predicted by the regression line. That is, we can observe $\rho(Y, \hat{Y})$. The model with the highest such correlation might then be deemed the ‘best-fitting’ model.

Week 4 through Week 6 Practice Exercises

1. Hill, Griffiths and Lim, Question 3.1. Instead of referring to Figure 2.9 in the textbook, you can refer to the Stata output given as:

Source	SS	df	MS	Number of obs = 40		
Model	190626.984	1	190626.984	F(1, 38) = 23.79		
Residual	304505.176	38	8013.2941	Prob > F = 0.0000		
Total	495132.16	39	12695.6964	R-squared = 0.3850		
				Adj R-squared = 0.3688		
				Root MSE = 89.517		

Food_exp	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
income	10.20964	2.093264	4.88	0.000	5.972052	14.44723
_cons	83.416	43.41016	1.92	0.062	- 4.463279	171.2953

2. Hill, Griffiths and Lim, Question 3.2.
3. A marketing agency surveyed movie-goers after they just viewed the latest blockbuster film. The participants were first divided into two age categories: (1) 25 yrs old or less, and (2) over 25 years old. Then, each participant was asked to rate the movie they just saw as either 1, 2, 3 or 4 stars. The results are as follows:

	Number of Stars Given			
Age	1	2	3	4
25 Yrs or less	5	6	9	10
Over 25 yrs	9	20	15	6

We hope to run a regression on the number of stars given as a function of X, where X is a dummy variable such that $X = 1$ if a person is over 25 years old and 0 otherwise.

- (a) Determine the mean number of stars given by each of the two groups.

- (b) Using only your information from part (a), determine the estimates of β_1 and β_2 in the linear regression model: Stars = $\beta_1 + \beta_2 X + \varepsilon$.
 - (c) Using the information from the above table, determine the sum of squared errors for the regression line of part (b).
 - (d) Using your results from the previous parts, determine if there is a significant difference, using $\alpha = 8\%$, in the mean number of stars given by each of the age groups.
- 4. Hill, Griffiths and Lim, Question 3.6.
 - 5. Hill, Griffiths and Lim, Question 4.1.
 - 6. Hill, Griffiths and Lim, Question 4.3. (Note that the notation $se(f)$ in part (b) represents the standard deviation used for a prediction interval.)
 - 7. Hill, Griffiths and Lim, Question 4.7.
 - 8. A nonlinear function form not explicitly discussed in the textbook is the reciprocal regression model:

$$Y = \beta_1 + \beta_2 \frac{1}{X} + \varepsilon$$

- (a) Determine both the slope and elasticity of the reciprocal regression model. Give a verbal interpretation of both calculations.
- (b) One potential economic application of a reciprocal regression is modeling the Phillips Curve, which measures year-to-year inflation rates (Y) as a function of annual unemployment percentage (X). Historically, low employment rates are often associated with high inflation rates and vice-versa (hence a reciprocal relationship).

Below are the annual inflation rates² and unemployment rates³ in the United States for $n = 5$ separate years:

² macro.trends.net/global-metrics/countries/USA/united-states/inflation-rate-cpi

³ [statista.com/statistics/193290/unemployment-rate-in-the-usa-since-1990/](https://www.statista.com/statistics/193290/unemployment-rate-in-the-usa-since-1990/)

Year	Unemployment Rate (X)	Inflation Rate (Y)
2005	5.1	3.39
2009	9.3	– 0.36
2014	6.2	1.62
2017	4.4	2.13
2022	3.6	8.00

Determine the regression line using the reciprocal regression model with inflation rate as a function of unemployment rate.

9. Hill, Griffiths and Lim, Question 4.8.

10. Hill, Griffiths and Lim, Question 4.9.

Week 4 through Week 6 Practice Exercises – Solutions

1. (a) The number of standard deviations for the interval is given by $TINV(.05, 38) = 2.024$. The other contributors to the interval are directly given in the Stata output. Ultimately, the confidence interval is given by:

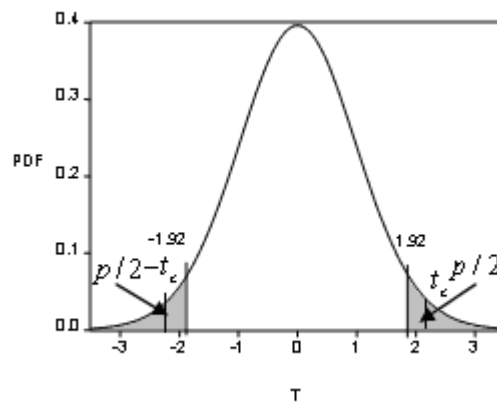
$$83.416 \pm (2.024)(43.41016) \Rightarrow -4.45 \text{ up to } 171.28$$

- (b) The hypothesis test is such that $H_0: \beta_1 = 0$, and $H_a: \beta_1 \neq 0$. The critical value is given by $TINV(.05, 38) = 2.024$.

The t-statistic is given by $\frac{83.416 - 0}{43.41016} = 1.92$.

Since $1.92 < 2.024$ we do not reject the null hypothesis.

- (c) The visual is given by:

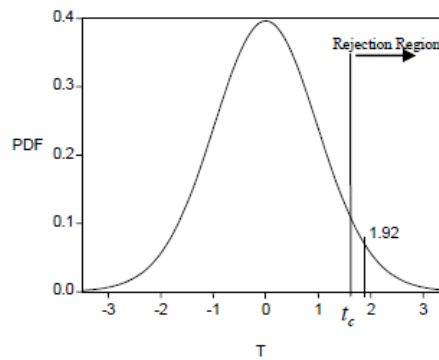


The p-value of 0.062 represents the accumulated probability in the two shaded regions. The $\alpha = 5\%$ level of significance is the accumulated probability within the two tails beyond the solid black lines. Since the former probability is greater than the latter, we do not reject the null hypothesis.

- (d) The hypothesis test is such that $H_0: \beta_1 \leq 0$, and $H_a: \beta_1 > 0$. The critical value is given by $TINV(.10, 38) = 1.686$.

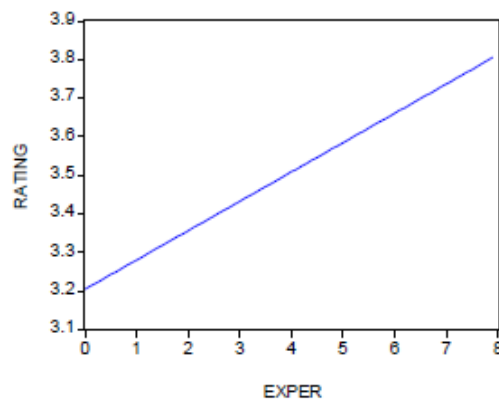
The t-statistic is (still) given by $\frac{83.416 - 0}{43.41016} = 1.92$.

Since $1.92 > 1.686$ we reject the null hypothesis. The associated p-value is $TDIST(1.92, 38, 1) = 3.12\%$. The associated visual is:



- (e) Overall, the level of significance is the total probability in the tails of the rejection region. By contrast, the confidence level is the total probability within the borderlines of the 'do not reject' region for a two-tailed hypothesis test. So, they are dual concepts. That is, for two-tailed hypothesis tests, confidence level = $1 - \text{level of significance}$.
- (f) False. The result of part (d) says that, if β_1 is truly equal to 0 then it is quite unlikely to have observed such a deviant value of β_1 in our sample data and, by rejecting the null, we are now willing to abandon this belief. The 5% represents the probability that we made a mistake in reversing our decision (i.e., the 5% represents $P(\text{Type I error})$). What we are actually 95% confident of is that β_1 falls in the range -4.45 up to 171.28 , which we determined in part (a).

2. (a) The sketch is given by:



The coefficient of EXPER indicates that, on average, a technical artist's quality rating goes up by 0.076 for every additional year of experience.

- (b) The number of standard deviations for the interval is given by $TINV(.05, 22) = 2.074$. Ultimately, the confidence interval is given by:

$$0.076 \pm (2.074) (0.044) \quad \Rightarrow \quad -0.015 \text{ up to } 0.167$$

- (c) The hypothesis test is such that $H_0: \beta_2 = 0$, and $H_a: \beta_2 \neq 0$. The critical value is given by $TINV(.05, 22) = 2.074$.

The t-statistic is given by $\frac{0.076 - 0}{0.044} = 1.73$.

Since $1.73 < 2.074$ we do not reject the null hypothesis.

- (d) The hypothesis test is such that $H_0: \beta_2 \leq 0$, and $H_a: \beta_2 > 0$. The critical value is given by $TINV(.10, 22) = 1.717$.

The t-statistic is (still) given by $\frac{0.076 - 0}{0.044} = 1.73$.

Since $1.73 > 1.717$ we reject the null hypothesis.

- (e) In general, $P(\text{Type I error}) = \alpha$ so, here, we are using a level of significance of $\alpha = 5\%$. Since the p-value of 9.82% is greater than the level of significance $\alpha = 5\%$ we *do not reject* the null hypothesis.

3. (a) For the 25 years old or younger crowd, we have:

$$E(\text{Stars} \mid 25 \text{ or less}) = \frac{5}{30}(1) + \frac{6}{30}(2) + \frac{9}{30}(3) + \frac{10}{30}(4) = 2.80$$

For the over 25 years old crowd, we have:

$$E(\text{Stars} \mid \text{over } 25) = \frac{9}{50}(1) + \frac{20}{50}(2) + \frac{15}{50}(3) + \frac{6}{50}(4) = 2.36$$

- (b) As discussed in class, b_1 = the sample mean of y-values for the observations assigned $X = 0$, so we have $b_1 = 2.80$. Further, b_2 = (the sample mean of y-values for the observations assigned $X = 1$) – (the sample mean of y-values for the observations assigned $X = 0$). Therefore, we have $b_2 = 2.36 - 2.80 = -0.44$.

- (c) Overall, the sum of squared errors is given by:

$$\begin{aligned} \sum_i \varepsilon_i^2 &= 5(1 - 2.80)^2 + 6(2 - 2.80)^2 + 9(3 - 2.80)^2 + 10(4 - 2.80)^2 + 9(1 - 2.36)^2 \\ &\quad + 20(2 - 2.36)^2 + 15(3 - 2.36)^2 + 6(4 - 2.36)^2 = 76.32 \end{aligned}$$

- (d) The hypothesis test is given by $H_0: \beta_2 = 0$ and $H_a: \beta_2 \neq 0$. Based on part (c), we immediately have:

$$MSE = \frac{1}{N-2} \sum_i \varepsilon_i^2 = \frac{1}{78} (76.32) = 0.9785$$

Since we are working with a dummy-variable regression, we have:

$$\text{Var}(b_2) = \frac{\text{MSE}}{n_{x=1}} + \frac{\text{MSE}}{n_{x=0}} = \frac{0.9785}{50} + \frac{0.9785}{30} = 0.0522$$

The t-statistic is therefore given by:

$$\text{t-statistic} = \frac{-0.44 - 0}{\sqrt{0.0522}} = -1.93.$$

and the critical value is given by $\text{TINV}(.08, 78) = 1.774$.

Since $-1.93 < -1.774$ we reject the null hypothesis and conclude that there is a significant difference between the population means at the $\alpha = 8\%$ level.

4. (a) The regression line is given by $\text{MOTEL_PCT} = 21.40 + 0.8646 \text{ COMP_PCT}$. Once the regression has been run, we can use the 'vce' command in Stata to find $\text{Var}(b_2) = 0.0411$.

The hypothesis test is such that $H_0: \beta_2 \leq 0$, and $H_a: \beta_2 > 0$. The critical value is given by $\text{TINV}(.02, 23) = 2.50$.

The t-statistic is given by $\frac{0.8646 - 0}{\sqrt{0.0411}} = 4.265$.

Since $4.265 > 2.50$ we reject the null hypothesis and conclude that the slope coefficient is significantly above zero at the $\alpha = 1\%$ level. That is, when the competitor's occupancy rate is high, our motel's occupancy rate is also high. The associated p-value is $\text{TDIST}(4.265, 23, 1) = 0.0145\%$.

- (b) The regression line is given by $\text{MOTEL_PCT} = 166.656 - 122.119 \text{ RELPRICE}$. Once the regression has been run, we can use the 'vce' command in Stata to find $\text{Var}(b_2) = 3404.75$.

The hypothesis test is such that $H_0: \beta_2 \geq 0$, and $H_a: \beta_2 < 0$. As in part (a), the critical value is given by $\text{TINV}(.02, 23) = 2.50$.

The t-statistic is given by $\frac{-122.119 - 0}{\sqrt{3404.75}} = -2.093$.

Since $-2.093 > -2.50$ we do not reject the null hypothesis at the $\alpha = 1\%$ level. That is, there is not sufficient evidence to conclude that a higher relative price compared to our competition deters potential customers from choosing our motel (although there would be sufficient evidence at a slightly higher level of significance, as suggested by demand theory). The associated p-value is $\text{TDIST}(2.093, 23, 1) = 2.38\%$.

- (c) The regression line is given by $\text{MOTEL_PCT} = 79.35 - 13.236 \text{ REPAIR}$. Once the regression has been run, we can use the 'vce' command in Stata to find $\text{Var}(b_2) = 35.5289$.

The hypothesis test is such that $H_0: \beta_2 \geq 0$, and $H_a: \beta_2 < 0$. The critical value is given by $\text{TINV}(.10, 23) = 1.714$. The logic behind this hypothesis test is that, since β_2 represents the change in occupancy rate during the repair period as opposed to otherwise, the alternative hypothesis asserts that occupancy rates were significantly down during the repairs.

The t-statistic is given by $\frac{-13.236 - 0}{\sqrt{35.5289}} = -2.22$.

Since $-2.22 < -1.714$ we reject the null hypothesis at the $\alpha = 5\%$ level. That is, there is ample evidence to conclude that occupancy rates were significantly lower during the repair period.

- (d) The number of standard deviations for the interval is given by $\text{TINV}(.05, 23) = 2.069$. Ultimately, the confidence interval is given by:

$$-13.236 \pm (2.069) (\sqrt{35.5289}) \Rightarrow -25.57 \text{ up to } -0.90$$

Overall, we are 95% confident that the true occupancy rate declined between 9% and 26% during the repair period. This is not a terribly precise estimate; we could improve its precision with a larger sample size.

- (e) The regression line is given by:

$$\text{MOTEL_PCT} - \text{COMP_PCT} = 16.861 - 14.118 \text{ REPAIR}.$$

Using the 'vce' command in Stata, we find $\text{Var}(b_2) = 15.8908$.

The hypothesis test is such that $H_0: \beta_2 \geq 0$, and $H_a: \beta_2 < 0$. The critical value is given by $\text{TINV}(.02, 23) = 2.50$. Here β_2 represents the decline in our motel's occupancy rate relative to our competitor during the repair period. That is, the alternative hypothesis asserts that our competitor was getting relatively more business during the repair period.

The t-statistic is given by $\frac{-14.118 - 0}{\sqrt{15.8908}} = -3.54$.

Since $-3.54 < -2.50$ we reject the null hypothesis at the $\alpha = 1\%$ level. That is, there is ample evidence to conclude that the difference in occupancy rates narrowed during the repair period.

- (f) The number of standard deviations for the interval is given by $TINV(.05, 23) = 2.069$. Ultimately, the confidence interval is given by:

$$-14.118 \pm (2.069) (\sqrt{15.8908}) \Rightarrow -22.37 \text{ up to } -5.87$$

Overall, we are 95% confident that the true difference in occupancy rates declined between 6% and 22% during the repair period. Although this is not a terribly precise estimate, it is reasonable to conclude that our motel's occupancy rate was adversely affected during the repair period as compared to our competitor.

5. (a) We are given $SST = 631.63$ and $SSE = 182.85$. Since $SST = SSR + SSE$, we have $SSR = 631.63 - 182.85 = 448.78$. Consequently, $R^2 = 448.78 / 631.63 = .7105$.

(b) We have:
$$\begin{aligned} \sum (y_i - \bar{y})^2 &= \sum (y_i^2 - 2y_i\bar{y} + \bar{y}^2) = \sum y_i^2 - 2\bar{y} \sum y_i + \sum \bar{y}^2 = \\ &= \sum y_i^2 - 2\bar{y} N\bar{y} + N\bar{y}^2 = \sum y_i^2 - 2N\bar{y}^2 + N\bar{y}^2 = \sum y_i^2 - N\bar{y}^2. \end{aligned}$$

Therefore, $SST = \sum (y_i - \bar{y})^2 = \sum y_i^2 - N\bar{y}^2 = 5930.94 - (20)(16.035)^2 = 788.5155$ and we consequently have $R^2 = 666.72 / 788.5155 = .8455$.

- (c) If $SSR / SST = .7911$ then $SSE / SST = 1 - .7911 = .2089$. So, $SSE = (.2089)(552.36) = 115.388$. We then have $\hat{\sigma}^2 = MSE = SSE / (N - 2) = 115.388 / 18 = 6.41$.

6. (a) We have:

$$E(XY) = \frac{1}{5} [(0)(6) + (1)(2) + (2)(3) + (3)(1) + (4)(0)] = 2.2$$

$$E(X) = \frac{1}{5} (0 + 1 + 2 + 3 + 4) = 2$$

$$E(Y) = \frac{1}{5} (6 + 2 + 3 + 1 + 0) = 2.4$$

$$\text{Var}(X) = \frac{1}{5} (0^2 + 1^2 + 2^2 + 3^2 + 4^2) - (2)^2 = 2$$

The regression line estimates are therefore:

$$b_2 = \frac{\text{Cov}(X, Y)}{\text{Var}(X)} = \frac{2.2 - (2)(2.4)}{2} = -1.3, \text{ and}$$

$$b_1 = 2.4 + (1.3)(2) = 5.$$

So, the predicted value of Y given X = 4 is $5 - (1.3)(4) = -0.20$.

(b) The sum of squared residuals is given by:

$$\sum_i \varepsilon_i^2 = (6 - 5)^2 + (2 - 3.7)^2 + (3 - 2.4)^2 + (1 - 1.1)^2 + (0 - (-0.2))^2 = 4.3$$

$$\text{which leads to } \text{MSE} = \frac{1}{N-2} \sum_i \varepsilon_i^2 = \frac{1}{3} (4.3) = 1.4333.$$

Since $E(X^2) = \frac{1}{5} (0^2 + 1^2 + 2^2 + 3^2 + 4^2) = 6$, we consequently have:

$$\text{Var}(b_1) = \text{MSE} \left[\frac{E(X^2)}{N \text{Var}_p(X)} \right] = (1.4333) \left(\frac{6}{(5)(2)} \right) = 0.86$$

$$\text{Var}(b_2) = \text{MSE} \left[\frac{1}{N \text{Var}_p(X)} \right] = (1.4333) \left(\frac{1}{(5)(2)} \right) = 0.14333$$

$$\text{Cov}(b_1, b_2) = -\bar{X} \text{Var}(b_2) = (-2)(0.14333) = -0.28666$$

In general, the variance for the prediction interval is given by:

$$\text{MSE} + \text{Var}(b_1) + X_0^2 \text{Var}(b_2) + 2 X_0 \text{Cov}(b_1, b_2)$$

which, at $X_0 = 4$, translates as:

$$1.4333 + 0.86 + (4)^2 (0.14333) + (2)(4)(-0.28666) = 2.29333$$

which implies $\text{se}(f) = \sqrt{2.29333} = 1.514$.

(c) The number of standard deviations for the prediction interval is given by $\text{TINV}(.05, 3) = 3.182$. Ultimately, the prediction interval is given by:

$$-0.20 \pm (3.182)(1.514) \Rightarrow -5.02 \text{ up to } 4.62$$

(d) When $X_0 = \bar{X}$, the general expression for the prediction interval becomes:

$$\text{MSE} + \text{Var}(b_1) + \bar{X}^2 \text{Var}(b_2) + 2 \bar{X} \text{Cov}(b_1, b_2)$$

$$\Rightarrow \text{MSE} + \text{Var}(b_1) + \bar{X}^2 \text{Var}(b_2) - 2 \bar{X}^2 \text{Var}(b_2)$$

$$\Rightarrow \text{MSE} + E(X^2) \text{Var}(b_2) - \bar{X}^2 \text{Var}(b_2)$$

$$\Rightarrow \text{MSE} + \text{Var}(b_2) [E(X^2) - \bar{X}^2]$$

$$\Rightarrow \text{MSE} + \text{Var}(b_2) [\text{Var}_p(X)]$$

$$\Rightarrow \text{MSE} + \text{MSE} \left[\frac{1}{N \text{Var}_p(X)} \right] [\text{Var}_p(X)]$$

$$\Rightarrow \text{MSE} + \text{MSE} \left[\frac{1}{N} \right]$$

$$\Rightarrow \text{MSE} \left[1 + \frac{1}{N} \right]$$

which, in this exercise, calculates as $(1.4333) \left[1 + \frac{1}{5} \right] = 1.72$.

We therefore have $\text{se}(f) = \sqrt{1.72} = 1.311$ and the prediction interval is given by:

$$2.40 \pm (3.182) (1.311) \Rightarrow -1.77 \text{ up to } 6.57$$

The width of the prediction interval in part (c) is $2 (3.182) (1.514) = 9.635$ whereas the width of the prediction interval here is $2 (3.182) (1.311) = 8.343$ which reinforces the fact that the width of the prediction interval lessens as X_0 approaches $\bar{X}$. This makes intuitive sense since, as a regression line varies, the $\hat{Y}$ -values corresponding to X -values near $X_0 = \bar{X}$ do not vary as much as the $\hat{Y}$ -values corresponding to X -values far away from $X_0 = \bar{X}$ (in either direction). In other words, $\text{Var}(\hat{Y})$ is lower the closer X_0 is to $\bar{X}$. Since $\text{Var}(\hat{Y})$ is the only component of the prediction interval variance that relies on the value of X_0 , we directly have that a prediction interval variance (and hence standard deviation) will decrease as X_0 approaches $\bar{X}$.

7. (a) In Assignment 2, Question 7 we found that the least squares estimator is:

$$b_2 = \frac{\sum XY}{\sum X^2}$$

and, when applying it to the data in Exercise 2.4, we obtain $b_2 = 176/91 = 1.934$. The least squares predictor of Y in this case is $\hat{Y} = 1.934 X$.

(b) We have $\text{SSE} = (4 - 1.934(1))^2 + (6 - 1.934(2))^2 + (7 - 1.934(3))^2 + (7 - 1.934(4))^2 + (9 - 1.934(5))^2 + (11 - 1.934(6))^2 = 11.604$.

Further, $\sum Y_i^2 = 16 + 36 + 49 + 49 + 81 + 121 = 352$.

Overall, we then have $R_u^2 = 1 - \frac{11.604}{352} = .9670$.

(c) We have $E(Y \hat{Y}) =$

$$\frac{1}{6}[(4)(1.934) + (6)(3.868) + (7)(5.802) + (7)(7.736) + (9)(9.67) + (11)(11.604)] = 56.731$$

as well as $E(Y) = 7.333$, $E(\hat{Y}) = 6.769$, $\sigma_Y^2 = 4.8889$ and $\sigma_{\hat{Y}}^2 = 10.9101$.

$$\text{Therefore, } \rho(Y, \hat{Y}) = \frac{56.731 - (7.333)(6.769)}{(\sqrt{4.8889})(\sqrt{10.9101})} = .971329, \text{ which implies}$$

$$r_{Y\hat{Y}}^2 = (.971329)^2 = .9435.$$

(d) We have $SST = (4 - 7.333)^2 + (6 - 7.333)^2 + (7 - 7.333)^2 + (7 - 7.333)^2 + (9 - 7.333)^2 + (11 - 7.333)^2 = 29.333$.

$$\text{We further have } SSR = (1.934 - 7.333)^2 + (3.868 - 7.333)^2 + (5.802 - 7.333)^2 + (7.736 - 7.333)^2 + (9.67 - 7.333)^2 + (11.604 - 7.333)^2 = 67.367.$$

Then, $SSR + SSE = 67.367 + 11.604 = 78.971 \neq 29.333 = SST$. Therefore, the decomposition *does not hold true* in this case.

8. (a) We have $dY = -\beta_2 \frac{1}{X^2} dX$, which implies that slope $= \frac{dY}{dX} = -\frac{\beta_2}{X^2}$. Therefore,

when X goes up by 1 unit, Y decreases by approximately $\frac{\beta_2}{X^2}$ units.

The elasticity is given by $\frac{dY/Y}{dX/X} = -\frac{\beta_2}{X^2} \frac{X}{Y} = -\frac{\beta_2}{XY}$. Therefore, when X goes

up by 1%, Y decreases by approximately $\frac{\beta_2}{XY}$ %.

(b) An expanded table of the sample data is:

Year	Unemployment Rate (X)	$\frac{1}{X}$	Inflation Rate (Y)	$(\frac{1}{X})^*(Y)$
2005	5.1	0.1961	3.39	0.6647
2009	9.3	0.1075	-0.36	-0.0387
2014	6.2	0.1613	1.62	0.2613
2017	4.4	0.2273	2.13	0.4841
2022	3.6	0.2778	8.00	2.2222

which leads to:

$$E\left(\frac{1}{X} Y\right) = \frac{1}{5}[0.6647 - 0.0387 + 0.2613 + 0.4841 + 2.2222] = 0.71872$$

$$E\left(\frac{1}{X}\right) = \frac{1}{5}(0.1961 + 0.1075 + 0.1613 + 0.2273 + 0.2778) = 0.194$$

$$E(Y) = \frac{1}{5}(3.39 - 0.36 + 1.62 + 2.13 + 8.00) = 2.956$$

$$\text{Var}\left(\frac{1}{X}\right) = \frac{1}{5}(0.1961^2 + 0.1075^2 + 0.1613^2 + 0.2273^2 + 0.2778^2) - (0.194)^2 = 0.00334$$

The regression line estimates are therefore:

$$b_2 = \frac{\text{Cov}\left(\frac{1}{X}, Y\right)}{\text{Var}\left(\frac{1}{X}\right)} = \frac{0.71872 - (0.194)(2.956)}{0.00334} = 43.49, \text{ and}$$

$$b_1 = 2.956 - (43.49)(0.194) = -5.481.$$

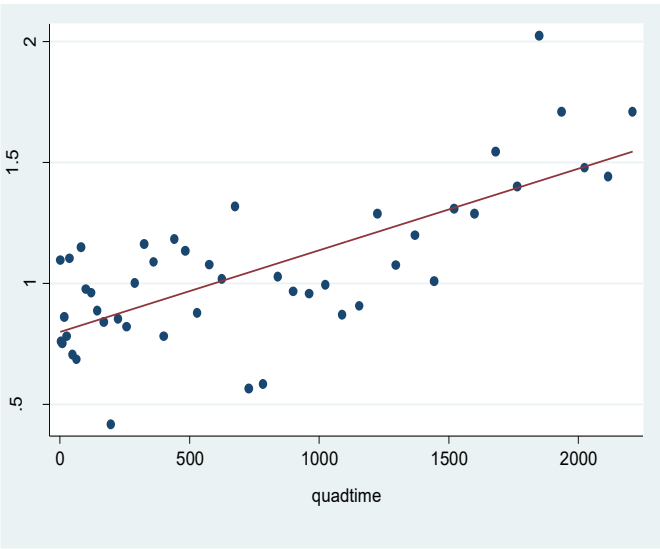
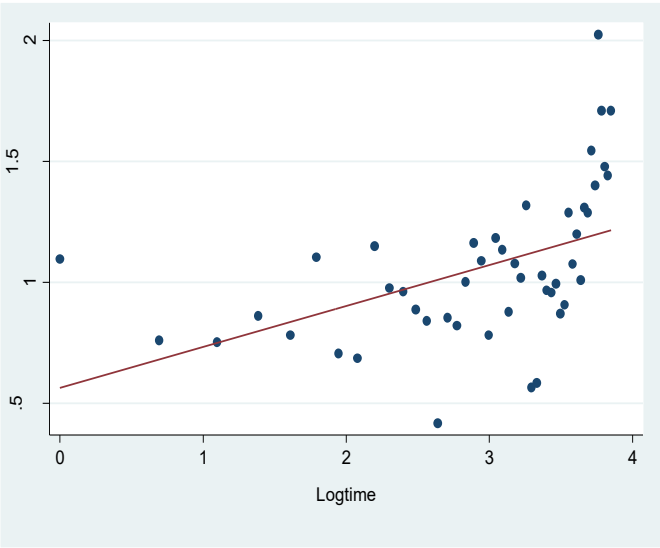
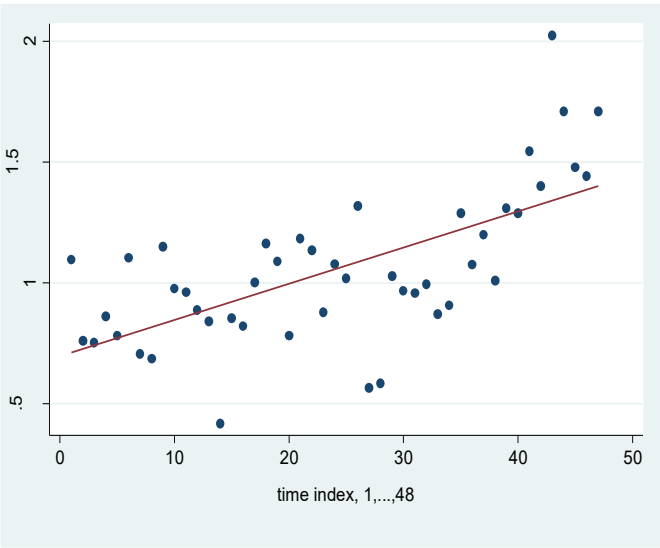
So, the estimated regression line is $\hat{Y} = 43.49 \frac{1}{X} - 5.481$.

9. (a) The linear regression leads to: $\text{Yield} = 0.6954 + 0.015 \text{ Time}$, with $R^2 = 42.45\%$

The linear-log regression leads to: $\text{Yield} = 0.5623 + 0.1696 \ln(\text{Time})$, with $R^2 = 22.54\%$

The quadratic regression leads to: $\text{Yield} = 0.7994 + 0.00034 (\text{Time})^2$, with $R^2 = 52.52\%$

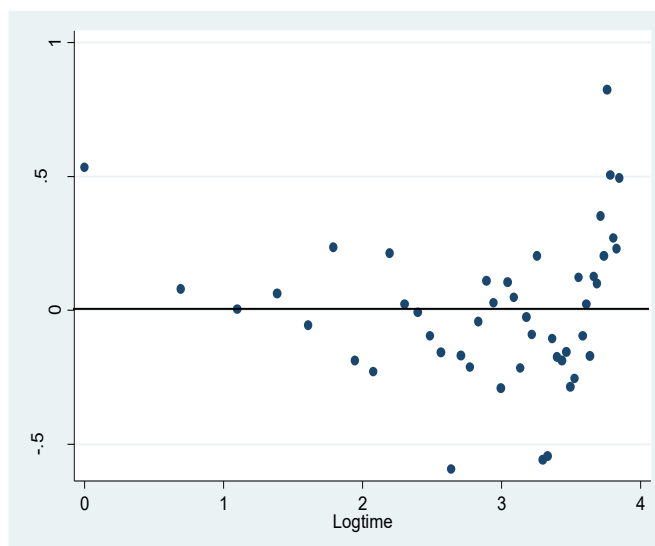
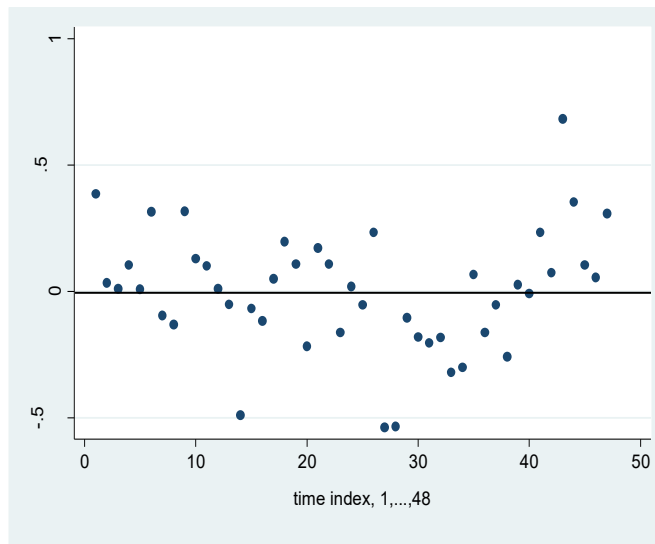
- (b) The plots of the fitted regression lines are shown below (linear model on top, then linear-log, then quadratic). In the first two graphs the scatter diagrams suggest a convex increasing functional form may work best. The last graph (with the quadratic functional form) appears to be the best fit.

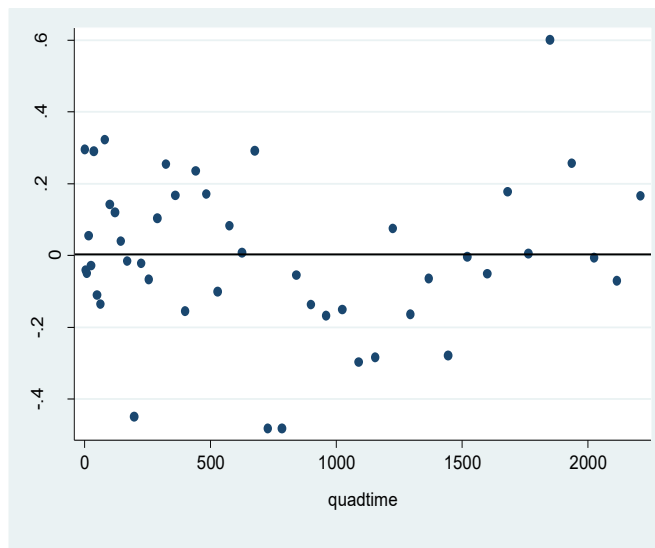


The plots of the residuals are shown below (linear model on top, then linear-log, then quadratic). For the linear model the residuals tend to have a pattern; we have positive residuals for extreme time periods and (mostly) negative residuals for intermediate time periods.

For the linear-log model the residuals seem to have higher variance for later time periods as opposed to earlier time periods. This contradicts the ‘constant variance’ or ‘homoskedasticity’ assumption of the simple linear regression model.

For the quadratic model the residuals still have some pattern to them, but not quite as pronounced as in the linear model. And, this residual plot is not nearly as heteroskedastic as the linear-log residual plot.

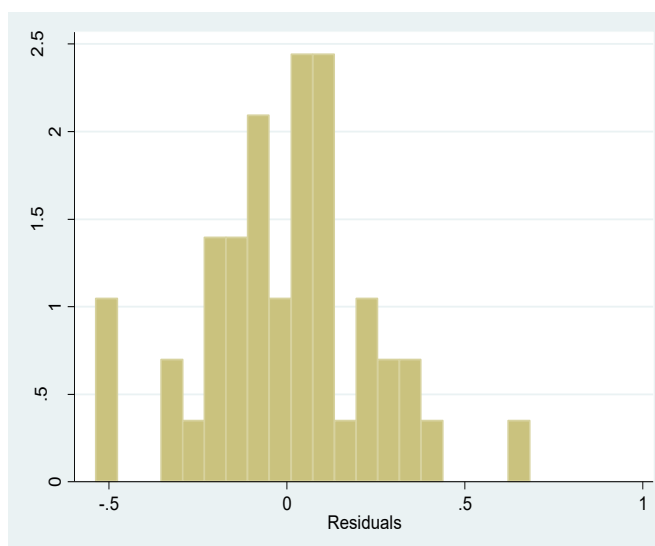


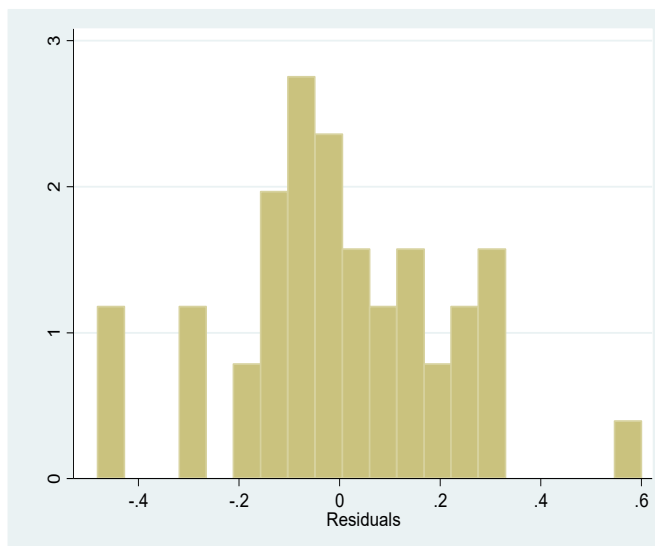
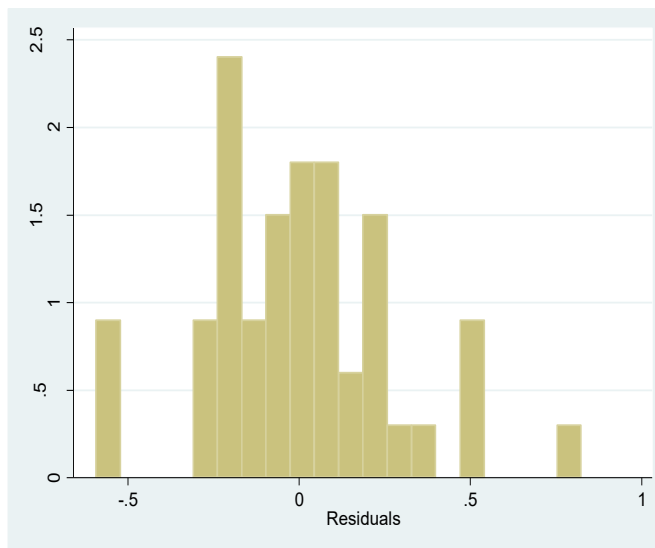


The residual distributions are shown below (linear model on top, then linear-log, then quadratic) using 20 bins on the x-axis. For the linear model the residuals look somewhat normally distributed although the low bars next to the modal bars are concerning, as is the high bar on the left tail.

For the linear-log model the residuals look more like a log-normal or chi-squared distribution. The mode seems displaced a bit from zero.

For the quadratic model the residuals look somewhat normally distributed although, like the linear model, the high bars toward the left tail are a bit concerning.





Overall, the quadratic model has the best visually-fitted plot, the highest R^2 -value and the best residual plot. It also arguably has the best residual distribution. Consequently, the quadratic model appears to be the best model among the three candidates for this sample data.

10. (a) Although the calculations can be done manually, the following sequence of commands can be used in Stata to automate the calculations:

The observations run from $t = 1$ to $t = 47$ (i.e., the data for $t = 48$ should be removed from the dataset from the outset since this is what the exercise hopes to predict. This can be done by typing the command 'drop in 48'). Then, in Stata, type:

- set obs 48 (tells Stata you want to include a 48th observation)
- replace time = 48 in 48 (sets time = 48 for the 48th observation. We do not set any other observation values; the others will be predicted)

Now, run the regression of interest and proceed by typing the following commands:

- predict ChapPred (gives the predicted values of the Chapman yield; we are arbitrarily labeling this variable 'ChapPred.' This will include a prediction for $t = 48$.)
- generate tvalue = invttail(45, .025) (gives the number of standard deviations from the t-table for a 95% confidence interval or prediction interval with 45 degrees of freedom)
- predict P1stddev, stdf (calculates the standard deviation for the prediction interval; we are arbitrarily labeling this variable 'P1stddev')
- generate lowerPI = ChapPred - tvalue*P1stddev (calculates the lower bound of the prediction interval; we are arbitrarily labeling this variable 'lowerPI')
- generate upperPI = ChapPred + tvalue*P1stddev (calculates the upper bound of the prediction interval; we are arbitrarily labeling this variable 'upperPI')

If you now look within your Data Editor there are several new columns in the spreadsheet with associated calculations that adhere to the above commands. We now just need to display the calculations corresponding to $t = 48$. This can be done as:

- list ChapPred lowerPI upperPI in 48 (lists the predicted value of the Chapman yield associated with $t = 48$ along with the lower and upper bounds of the prediction interval)

When the above steps are conducted on the linear model, the predicted value is 1.4166 while the prediction interval extends from 0.9071 up to 1.9260 (the standard deviation for the prediction interval is .2529).

When the above steps are conducted on the linear-log model the predicted value is 1.2189 while the prediction interval extends from 0.6391 up to 1.7987 (the standard deviation for the prediction interval is .2879).

When the above steps are conducted on the quadratic model the predicted value is 1.5771 while the prediction interval extends from 1.1047 up to 2.0495 (the standard deviation for the prediction interval is .2345).

In actuality, the Chapman yield associated with $t = 48$ is 1.8435. This value is included within the prediction interval bounds of both the linear and quadratic models but not within the bounds of the linear-log model.

- (b) The slope of the linear model is $\beta_2 = 0.015$. Since this is constant for any t , the estimated slope at $t = 48$ is 0.015.

The slope of the linear-log model is $\frac{\beta_2}{t}$. At $t = 48$, the slope is $\frac{0.1696}{48} = 0.00353$.

The slope of the quadratic model is $2 \beta_2 t$. At $t = 48$, the slope is $2 (.00034) (48) = 0.0326$.

- (c) In general, the elasticity is given by $(\text{slope}) \times \frac{t}{y}$. We therefore have:

The elasticity of the linear model at $t = 48$ is $(0.015) (48 / 1.4166) = .5083$

The elasticity of the linear-log model at $t = 48$ is $(0.00353) (48 / 1.2189) = .1390$

The elasticity of the quadratic model at $t = 48$ is $(0.0326) (48 / 1.5771) = .9922$

- (d) The slope calculations in part (b) estimate the marginal *unit-change* in y -values for a one-year increase in time.

The elasticity calculations in part (c) estimate the *percentage* change in y -values for a 1% increase in time.